



TIME SERIES ANALYSIS

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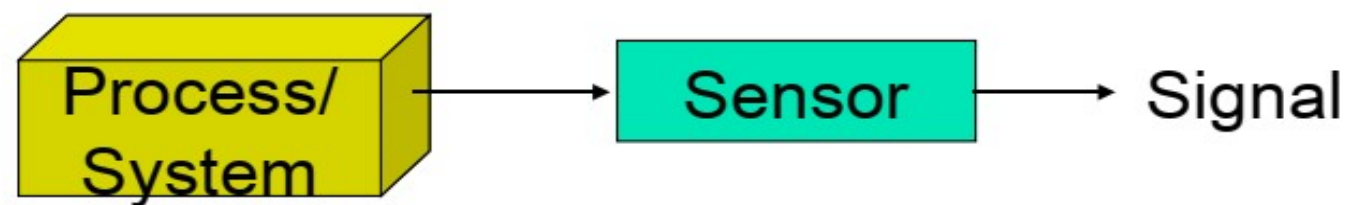
Presentation Contents



- ▶ Signals and Systems
- ▶ Analog and discrete signals
- ▶ Discrete signals and systems
- ▶ Operations on discrete signals
- ▶ Discrete system classification
- ▶ Properties of discrete systems
- ▶ Linear time invariant systems
- ▶ Difference equations

What is a signal?

- A signal is data that carries some information about the system from which it originates.
 - *It is an information carrier. It cannot exist without a system.*
 - *Signals are generally a function of **time**, **frequency** or **space**.*



E.g.:

- Heat exchanger
- Human body
- Atmosphere
- Earth
- Planet
- Stock Market

- Usually h/w
- External or internal (e.g., human body)



E.g.:

- Temperature
- Blood pressure
- Cloud movement
- Seismic vibration
- Satellite Image
- Stock prices

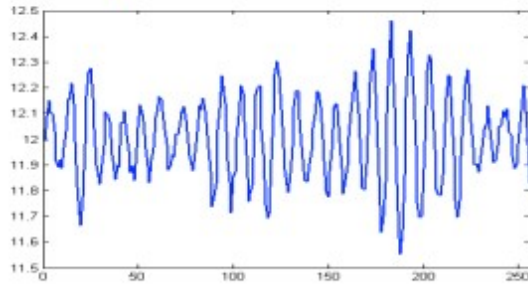
Representation of signals

- Any signal is a measured quantity (or an observed quantity) and consists of two pieces of information: the **amplitude** or the **value** and **independent variable** or the **domain** variable.
 - ▶ The values could be continuous or discrete; the independent variable is typically one or more of time, space, frequency
- Signal whose domain is continuous, are represented by continuous lines
- Discrete-time (domain) signals on the other hand are typically represented by what are known as *stem* plots
 - ▶ If the sample size is large or the grid is too closely-spaced, then one generally connects one value to the other using a continuous line
- Note: The choice of a particular type of representation in a graphical way depends on the application and can play a very important role in making any inferences

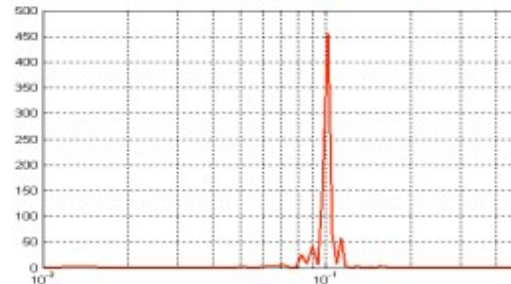
Examples of signals

- The independent quantities (time, frequency or space) are known as dimensions.
- Signals can be 1-dimensional, 2-D, 3-D or more.

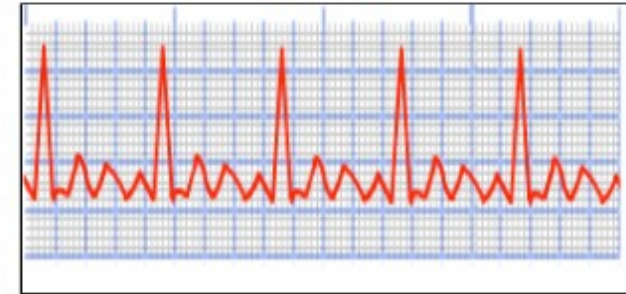
Valve output from a control loop as a function of time



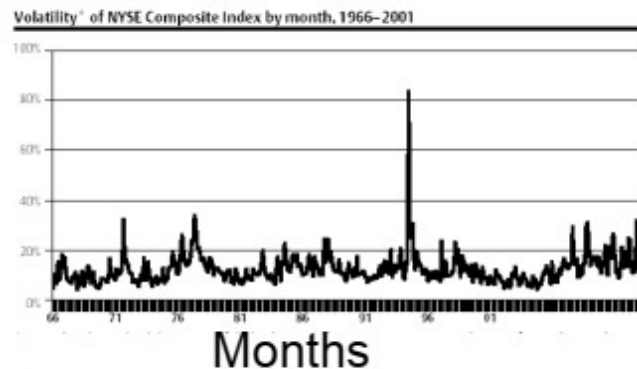
Valve output (power) as a function of frequency



ELECTROCARDIOGRAM (ECG)



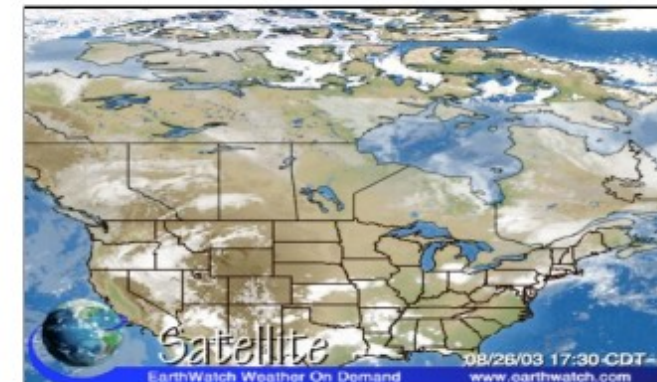
Volatility of NYSE Composite Index from 1966-2001



2-D Image of Mars on June 04, 2003



3-D Satellite Image of Canadian weather condition



Signal Analysis

- ▶ Signals carry information
 - ▶ A signal which does not carry information or carries information not desired is known as noise/noisy signal
- ▶ Aim of signal analysis
 - ▶ Extract useful information carried by the signal to suit the application
- ▶ Methods
 - ▶ The methods for signal analysis will depend on the type of the signal and nature of the information being carried by the signal
 - ▶ There are some common methodologies and some specific ones for specific signals

Classification of signals

- ▶ Signals can be classified into various types by
 - ▶ Nature of the independent variables
 - ▶ Value of the function defining the signals
- ▶ Examples:
 - ▶ Discrete/continuous function
 - ▶ Discrete/continuous independent variable
 - ▶ Real/complex valued function
 - ▶ Scalar (single channel)/Vector (multi-channels)
 - ▶ Single/Multi-trial (repeated recordings)
 - ▶ Dimensionality based on the number of independent variables (1D/2D/3D)
 - ▶ Deterministic/random
 - ▶ Periodic/aperiodic
 - ▶ Even/odd
 - ▶ Many more....

Classification - Discrete/continuous signals

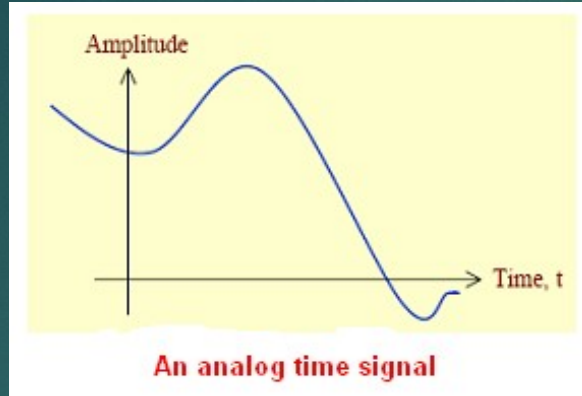
- ▶ Normally, the independent variable is time
- ▶ Continuous time signal
 - ▶ Time is continuous
 - ▶ Defined at every instant of time
- ▶ Discrete time signal
 - ▶ Time is discrete
 - ▶ Defined at discrete instants of time - it is a sequence of numbers
- ▶ Four classifications based on time/amplitude - continuous/discrete:
 - ▶ Analogue, digital, sampled, quantised boxcar

Classification - Discrete/continuous signals (cont)

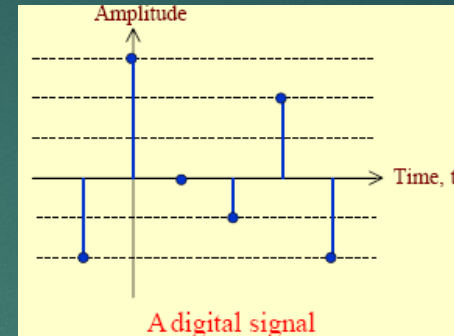
- ▶ Analogue signal
 - ▶ Continuous time signal with continuous amplitude, eg. music stored on cassette tape.
- ▶ Digital signal
 - ▶ Discrete time signal with discrete valued amplitudes represented by a finite number of digits, eg. music stored on hard disk.
- ▶ Sampled data signal
 - ▶ Discrete time signal with continuous valued amplitudes (i.e. amplitude can take any value)
 - ▶ Digital signal is thus quantised sampled data signal
- ▶ Quantised boxcar signal
 - ▶ Continuous time signal with discrete valued amplitudes

Classification - Discrete/continuous signals (cont)

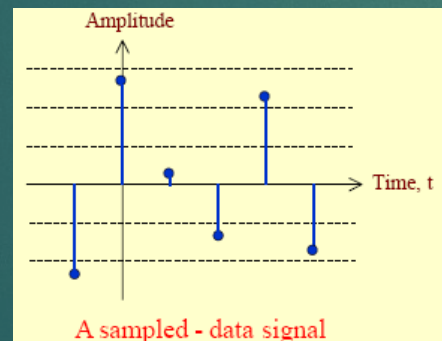
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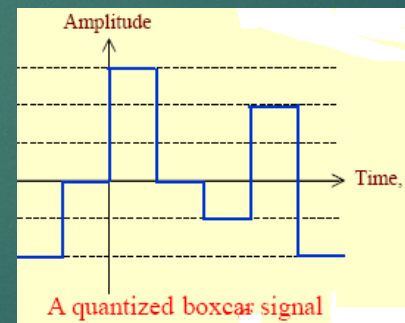
Amplitude- continuous
Time-continuous



Amplitude- discrete
Time-discrete



Amplitude- continuous
Time-discrete



Amplitude- discrete
Time-continuous

Figures from S.K.Mitra, DSP 3e

Random vs deterministic signal

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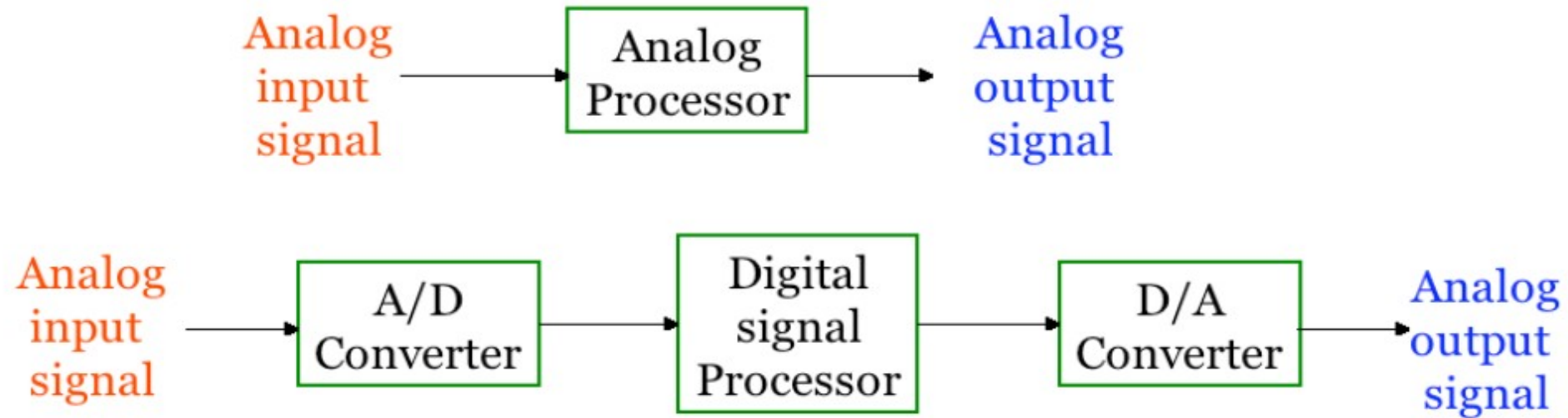
- ▶ **Deterministic signal**

- ▶ A signal that can be predicted using some methods like a mathematical expression or look-up table
- ▶ Easier to analyse

- ▶ **Random (stochastic)**

- ▶ A signal that is generated randomly and cannot be predicted ahead of time
- ▶ Most biological signals fall in this category
- ▶ More difficult to analyse

Elements of DSP



- ◆ Digital signal processing consists of mainly 3 elements:
 - A/D Converter: Comprises sampling, quantization and coding.
 - Digital Signal Processor: Typically a computer or digital hardware.
 - D/A Converter: Comprises an interpolator or an approximator.
- ◆ Sampling does not result in loss of information or distortion.
- ◆ Quantization results in some loss depending on the accuracy, measured by the number of bits.

Time domain operations

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▶ **Scaling**

- ▶ Multiplication of a signal by a constant, α
- ▶ Amplification if the α (gain) >1
- ▶ Attenuation if the $\alpha <1$
- ▶ Eg.: $y(t)=\alpha x(t)$

▶ **Delay/advance**

- ▶ Delays or advances the signal, $y(t)=x(t)$ by a certain time, t_0
- ▶ Eg.: Delay, $y(t)=x(t-t_0)$; Advance, $y(t)=x(t+t_0)$

▶ **Addition/subtraction**

- ▶ Addition/subtraction of signals to obtain a combined signal
- ▶ Eg.: $y(t)=x_1(t)+x_2(t)+x_3(t)$

Time domain operations (cont)

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- ▶ **Product**

- ▶ Product of two or more signals
- ▶ Eg.: $y(t) = x_1(t) \cdot x_2(t)$

- ▶ **Differentiation/Integration**

- ▶ Differentiation/Integration of a signal to produce a new signal
- ▶ Eg:

$$y(t) = \int_{-\infty}^t x(t) dt \qquad y(t) = \frac{dx(t)}{dt}$$

- ▶ **Combination of operators** $y(t) = x_1(t) + x_2(t - t_0) \cdot x_3(t) + \alpha x_4(t)$

- ▶ It is common to combine operators to generate a new signal
- ▶ Eg.:

- ▶ **Analogue/discrete-time operators**

- ▶ Scaling, addition/subtraction, delay, product – implemented in both analog and discrete-time signals
- ▶ Differentiation/integration – implemented in analog signals, only an approximation can be implemented with discrete-time signals

1D time series – some mathematical notations

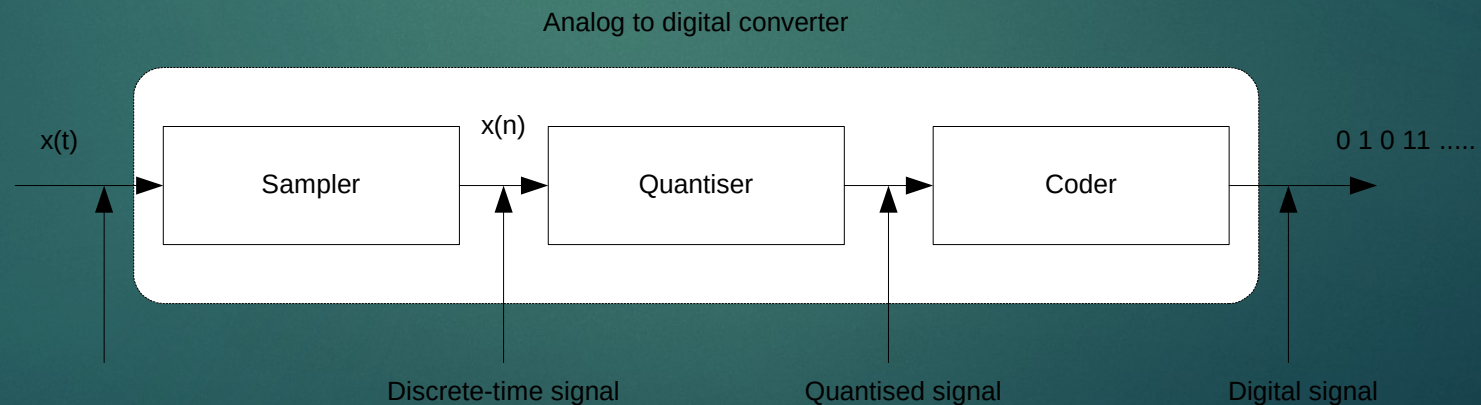
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- ▶ A 1D time series
 - ▶ $y=f(t)$ for continuous independent variable time
 - ▶ $y=f(n)$ for discrete independent variable n
 - ▶ Every value of $f(n)$ is called a sample
 - ▶ Discrete-time signal can be generated by sampling a parent continuous-time signal at uniform intervals of time
 - ▶ Then, discrete variable n can be normalised to assume integer values as a representation of t .

Analogue to digital conversion

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- ▶ Recorded biological signals are analogue signals
- ▶ To process analogue signals by digital means (like using a PC), we need to convert them to digital form
- ▶ I.e. to convert them to a sequence of numbers having finite precision
- ▶ This process is known as analogue to digital (A/D) conversion, which involves
 - ▶ i) Sampling
 - ▶ ii) Quantisation
 - ▶ iii) Coding



Sampled-data system



A typical sampled-data system comprising a zero-order hold, the continuous-time process and a sampler.

The need to sample data arose for various reasons. and gained momentum with the advent of digital process control in the 1960s.

The key elements of a sampled data system are:

- 1 Sampler, S : Obtains samples of the continuous-time signal

(output) at specified times (regular/irregular). In general, an A-to-D converter is used.

- 1 Zero-Order Hold, ZOH or H : Holds the discrete-signal constant over one sample period before being sent to the actuator. In general, a D-to-A converter is used.

- 2 Process, G : The process of interest.

Sampling: Concepts

Sampling is the act of obtaining the values of a continuous-time signal $x(t)$ at a set of discrete points. Define t_k , $k \in \mathbb{Z}$ as the k^{th} sampling instant. Then the discrete-time signal is the sequence $\{x(t_k)\}$.

If $t_k = kT_s$, where T_s is the sampling interval (period), then it is known as periodic sampling. Then $F_s = 1/T_s$ is known as the sampling rate or sampling frequency. In such a case, the value of the discrete-time signal at the k^{th} instant is $x(kT_s)$ or simply $x[k]$ (or $x[n]$).

Notation: We shall use square brackets to indicate a discrete-time signal henceforth.

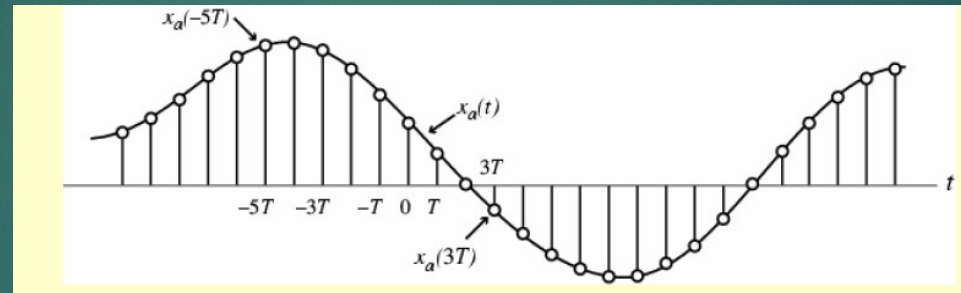
Q: What are the units of T_s and F_s ?

The units of T_s are time units/sample (we usually ignore the unit /sample). Likewise, the units of F_s are samples/time. Therefore a value of $F_s = 50$ Hz means 50 samples are obtained in one second. We can also express frequency in radians/time by merely multiplying F_s by 2π , $\omega_s = 2\pi F_s$.

Throughout the course, we shall only deal with periodic sampling.

The Sampling Process

- Often, a discrete time sequence $x[n]$ is developed by uniformly sampling an analogue signal $x(t)$ as indicated below
- I.e. conversion of a continuous-time, continuous amplitude signal into discrete-time signal (i.e. continuous-amplitude, discrete-time)
- Done by taking samples at specific uniform intervals of time
- The sampling interval is the time of one of these uniform intervals
- Sampling frequency is the number of uniform time intervals in one second



- The relation between the two signals are

$$x[n] = x_a(t)|_{t=nT} = x_a(nT), n = \dots, -2, -1, 0, 1, 2, \dots$$

Discrete-time
signal

Analogue signal

Concept of frequency

- Frequency relates to the periodicity of signals:
 - ♦ *Has the dimensions of inverse of time.*
- A simple continuous-time oscillatory signal:

$$x_a(t) = A \cos(\Omega t + \theta) \quad \Omega = 2\pi F$$

Amplitude *Frequency (radians/sec)* *Phase (radians)* *Frequency (cycles/sec)*

- Discrete-time sinusoid as a function of sample no. 'n':

$$x_d(n) = A \cos(\omega n + \theta) \quad \omega = 2\pi f$$

Amplitude *Frequency (radians/sample)* *Phase (radians)* *Frequency (cycles/sample)*

Frequency of a Sampled Signal

If $x(t) = A \sin(2\pi Ft)$ is sampled at a frequency F_s , what is the frequency of the resulting sampled signal $x[n]$?

$x(t) = A \sin(2\pi Ft) \implies x[n] = x(nT_s) = A \sin(2\pi FnT_s) = A \sin(2\pi Fn \frac{1}{F_s} t)$. Thus the frequency of a discrete-time sampled signal obtained by sampling a continuous-time sinusoid of frequency F at a sampling rate F_s is

$$f = \frac{F}{F_s}$$

. For example, a continuous-time sinusoid of frequency $F = 50$ Hz sampled at $F_s = 150$ samples/sec yields a discrete-time signal with a frequency $f = F/F_s = 50/150 = 0.3$ cycles/sample.

Note: It is useful to remember the units of frequencies of signals in both domains as well as that of the sampling frequency.

Exercise: Compare the frequencies of a discrete-time signal obtained by sampling a continuous-time sinusoid of frequency $F = 50$ Hz sampled at two different frequencies $F_{s,1} = 200$ samples/sec and $F_{s,2} = 40$ samples/sec.

Frequency in discrete-time

- A discrete-time sinusoid is periodic iff f is rational
- Discrete-time sinusoids with a frequency separation of $2k\pi$ are identical (k is an integer)

- ▶ This is due to the fact that period of a sinusoid is 2π

$$\cos((\omega_0 + 2\pi)n + \theta) = \cos(\omega_0 n + 2\pi n + \theta) = \cos(\omega_0 n + \theta)$$

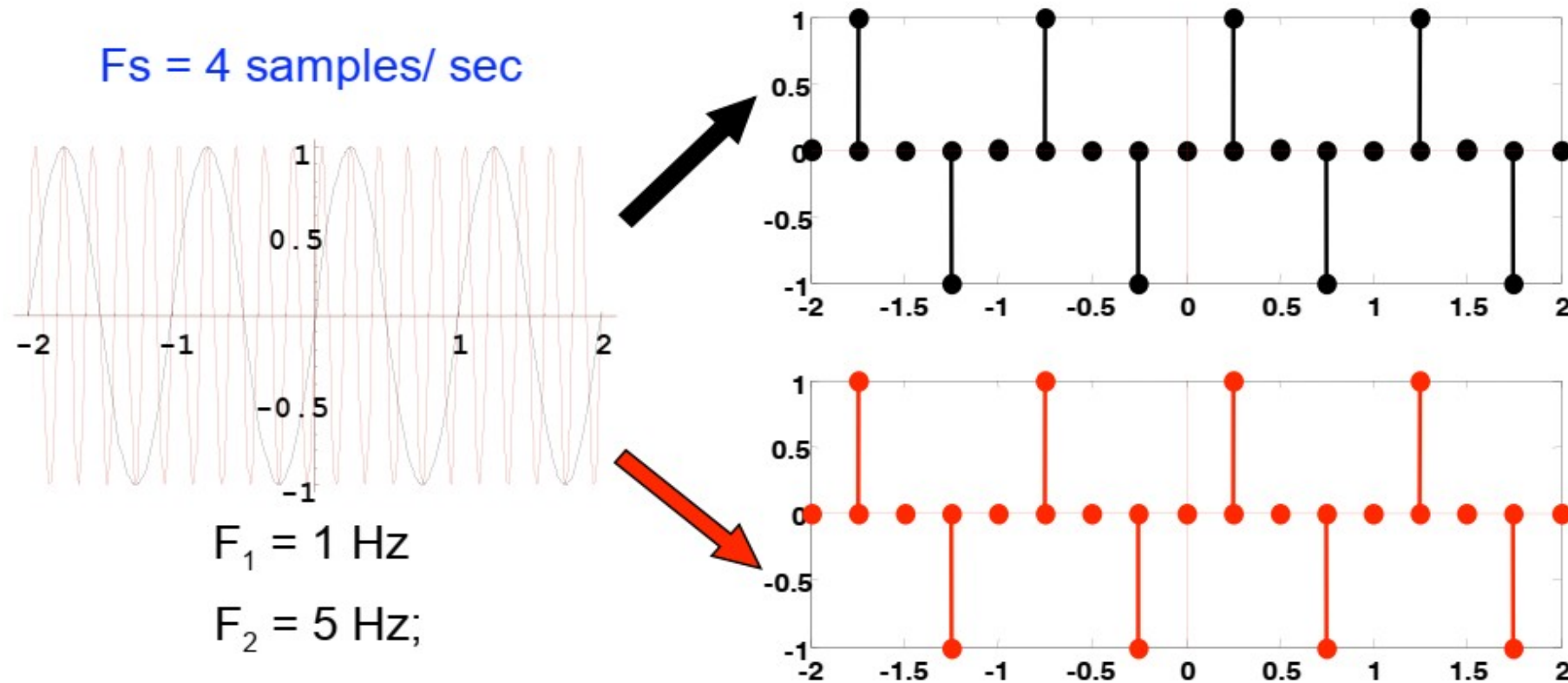
- ▶ All sinusoids with frequencies $\omega_k = \omega_0 + 2k\pi$ are indistinguishable
- ▶ Sinusoids with frequencies $-\pi < \omega_0 < \pi$ are distinct
- ▶ A sinusoid with frequency $|\omega_0| > \pi$ **will have an alias** of a sinusoid with another frequency $|\omega_1| < \pi$
- The maximum frequency of oscillation in a discrete-time sinusoid is $\omega = \pi$
 - ▶ All higher frequencies get mapped back into the range $-\pi < \omega_0 < \pi$

Aliasing

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- Aliasing
 - Aliasing causes ambiguities in reconstruction, i.e. distorts the sampled signal
 - To avoid aliasing, the sampling frequency has to be more than twice of the highest frequency contained in $x(t)$
 - This is known as Nyquist theorem and the minimum frequency known as Nyquist frequency(rate)

Aliasing Example



- Two different C.T. sinusoids yield same $x[n]$ when sampling frequency is not high enough.
- In general, C.T. signals with frequencies $F \pm kF_s$ are aliases of $x(t)$ with frequency F at sampling frequency F_s .

Sampling Theorem

Definition

A continuous-time signal with frequency F Hz has to be sampled at least at twice this frequency, *i.e.*, $F_{s,min} = 2F$

- If the signal contains more than one frequency, then the maximum frequency determines the minimum sampling frequency
- Typically a sampling rate of 3-4 times the maximum frequency is implemented in practice
- In general, it is not possible to know the maximum frequency *a priori*. Therefore, the analog signal is filtered such that the maximum frequency is clipped to a desired value which is then used for sampling.
 - ▶ Such a filter is known as the **anti-aliasing** filter

Example – Nyquist rate computation

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- Consider the analog signal
 - $x(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$
 - What is the Nyquist frequency (rate)?

- Answer
- Use the generic term, $A \cos 2\pi f t$ (or $A \sin 2\pi f t$) to compute the frequencies present in the signal $\Rightarrow$ which are 25 Hz, 150 Hz and 50 Hz
- So, Nyquist frequency is $2 \times$ highest frequency = $2 \times 150 \text{ Hz} = 300 \text{ Hz}$
- In practise, we normally sample at a much higher rate than Nyquist frequency

Sample Problem in Sampling

Consider the analog signal $x(t) = 3 \cos(100\pi t)$

- 1 Determine the minimum sampling rate required to avoid aliasing
- 2 Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?
- 3 Suppose that the signal is sampled at the rate $F_s = 75$ Hz.. What is the discrete-time signal obtained after sampling?
- 4 What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtained in part (c)?

Reconstruction and Zero-order Hold

The inversion of the sampling operation *i.e.*, converting the sequence of numbers $x[n]$ to a continuous-time signal $x(t)$ is known as *reconstruction*.

In computer-controlled systems as well as in many other systems, it becomes necessary to convert the control actions into continuous-time signals that can be sent to the actuators.

There are different types of reconstruction depending on the reconstruction device that is used.

❶ **Shannon Reconstruction:** The Shannon reconstruction is given by:

$$f(t) = \sum_{k=-\infty}^{k=\infty} f(kh) \sin c \frac{\omega_s(t - kh)}{2}.$$

This is an ideal reconstructor. It requires past and future samples and is therefore non-causal. It introduces a delay in the reconstruction. Due to the delay, it is not useful to control applications. However, it is used in communications and signal processing.

ZOH

- ① Zero-Order Hold (ZOH): This is probably one of the most commonly used reconstruction

devices. The reconstruction is given by:

$$f(t) = f(kh) \quad t_k \leq t < t_{k+1}.$$

The old value is held constant until the new conversion is ordered. The zero-order hold has several advantages over the Shannon's reconstructor. Also, it can be used for non-periodic sampling.

The ZOH is exact for signals that are piecewise constant over the sampling interval (such as step signals). For all other signals, there is a certain reconstruction error involved. The largest value of the error is given by:

$$e_{ZOH} = \max_k |f(t_{k+1}) - f(t_k)| \leq h \max_t |f'(t)|$$

Quantisation and Coding

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- ▶ Quantisation
 - ▶ This is the conversion of discrete-time continuous amplitude signal (after sampling) into discrete-time discrete amplitude signal
- ▶ Quantisation levels
 - ▶ The value of each sample is represented by a value selected from a finite set of possible values
 - ▶ Quantisation width = $(x_{\max} - x_{\min}) / (\text{No of levels} - 1)$
 - ▶ Quantisation error is the error in the quantisation process by rounding to the nearest level
- ▶ Coding
 - ▶ In the coding process, each discrete value is represented by a certain number of bits
 - ▶ $2^{\text{number of bits}} \geq \text{no of levels}$

Example of performing quantisation and coding

- Example of quantisation and coding
- Eg: Assume after sampling, we have the 5th sample, $x(5)=1.66$ volts
 - Assume 9 levels are used to represent amplitude values of 0 V (min) to 2 V (max)
 - Quantisation width = $(x_{\max}-x_{\min})/(\text{No of levels} - 1) = 0.25$ V
 - The possible quantised values are 0, 0.25, 0.5, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00
 - So, $x(5) = 1.75$ volts (rounded to nearest level) = 7th quantised level
 - $2^{\text{number of bits}} \geq \text{no of levels}$, so we need at least 4 bits
 - After coding, it is 0111
- Quantisation error
 - 1.66 V is represented as 1.75 V, so there is an error of 0.09 V, this is the quantisation error
 - This error can be reduced by using more bits in coding thereby increasing the number of quantisation levels
 - In some hardware, the quantisation is done to the nearest lower level, so it would be 1.50 instead of 1.75

Quantisation and coding – further example

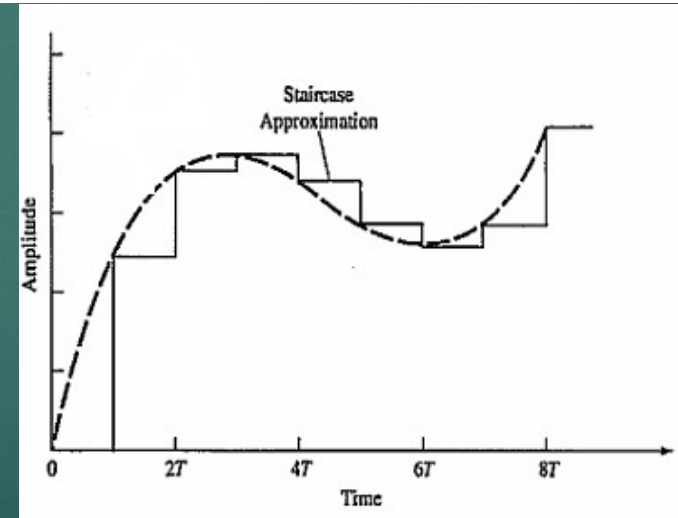
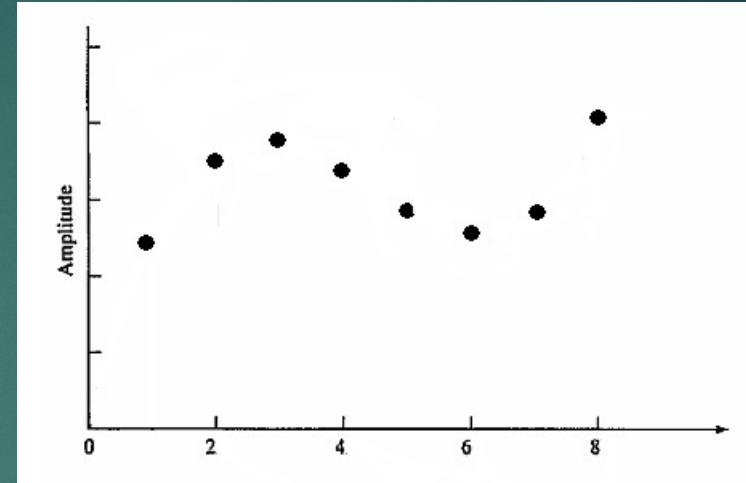
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- ▶ For the same example in the previous, what would be the code and quantisation error if 17 levels were used?
 - ▶ 17 levels in the range 0 V (min) to 2 V (max)
 - ▶ Quantisation width = $(x_{\max} - x_{\min}) / (\text{No of levels} - 1) = 0.125 \text{ V}$
 - ▶ The possible quantised values are 0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.00, 1.125, 1.25, 1.375, 1.50, 1.625, 1.75, 1.875, 2.00
 - ▶ So, $x(5) = 1.625$ volts (rounded to nearest level) = 13th quantised level
 - ▶ $2^{\text{number of bits}} \geq \text{no of levels}$, so we need at least 5 bits
 - ▶ After coding, it is 01101
- ▶ Quantisation error
 - ▶ 1.66 V is represented as 1.625 V, so there is an error of 0.035 V, this is the quantisation error
 - ▶ This error is reduced by the increase of bits/quantisation levels

Digital to analog (D/A) conversion

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- ▶ This is not important for this course
- ▶ Converts digital to analogue signals by performing some kind of interpolation to 'connect the dots'
- ▶ Eg: zero-order hold/staircase approximation A/D converter (the simplest)



Sequences

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- Sometimes, a discrete-time signal is known as a sequence and vice versa
- A discrete-time signal may be a finite length or an infinite-length sequence
 - Eg: $x[n]=3+4n^3$, $-5 \leq n \leq 4$ is a finite length sequence with length $4-(-5)+1=10$
 - $x[n]=\sin(0.1n)$ is an infinite-length sequence
- An N length sequence can be increased by padding with zeros
 - Eg: length 3 changed to length 5

$$x[n] = n^3, \quad 1 \leq n < 4$$

$$x_{pad}[n] = \begin{cases} n^3, & 1 \leq n < 4 \\ 0, & 4 \leq n \leq 5 \end{cases}$$

Discrete-Time Signals and Systems

- ▶ Introduction
- ▶ Discrete-Time Signals: Sequences
- ▶ Discrete-Time Systems
- ▶ Linear Time-Invariant (LTI) Systems
- ▶ Properties of LTI Systems
- ▶ Linear Constant-Coefficient Difference Equations

Discrete-Time Signals: Sequences

- ▶ Discrete-Time signals are represented as

$$x = \{x[n]\}, \quad -\infty < n < \infty, \quad n : \text{integer}$$

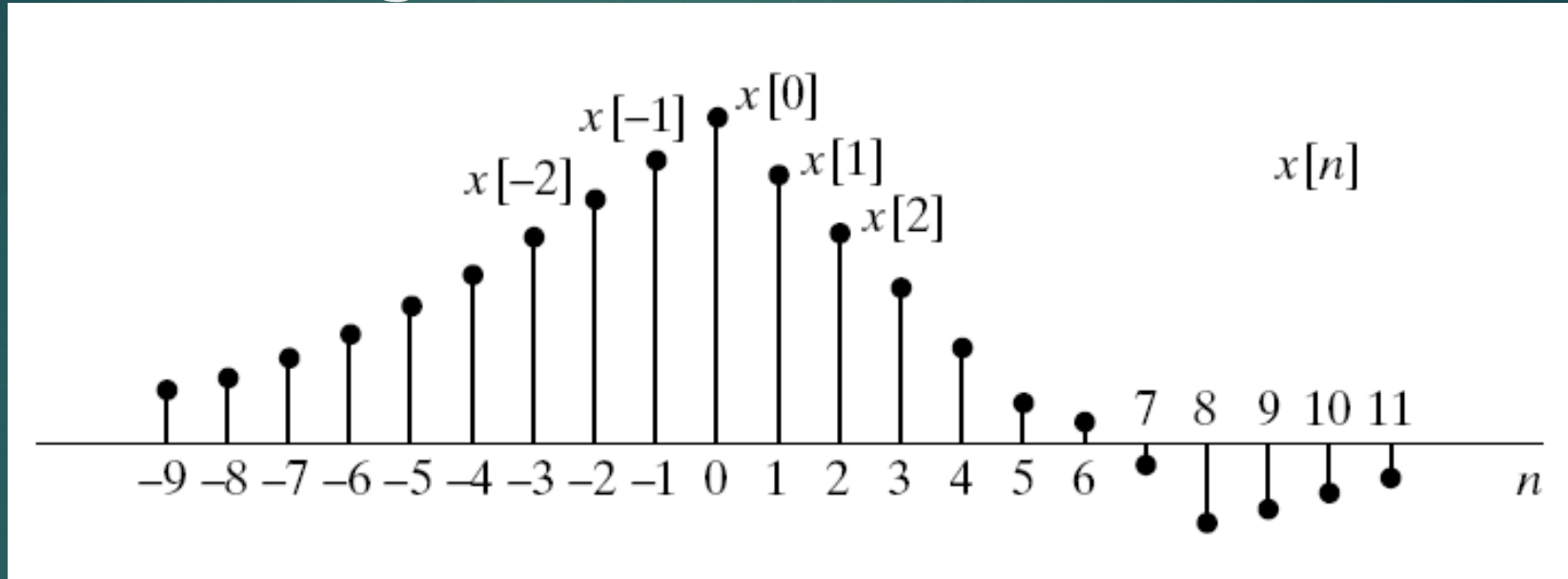
Cumbersome, so just use $x[n]$

- ▶ In sampling,

$$x[n] = x_a(nT), \quad T : \text{sampling period}$$

- ▶ $1/T$ (reciprocal of T) : sampling frequency

Graphical representation of a discrete-time signal



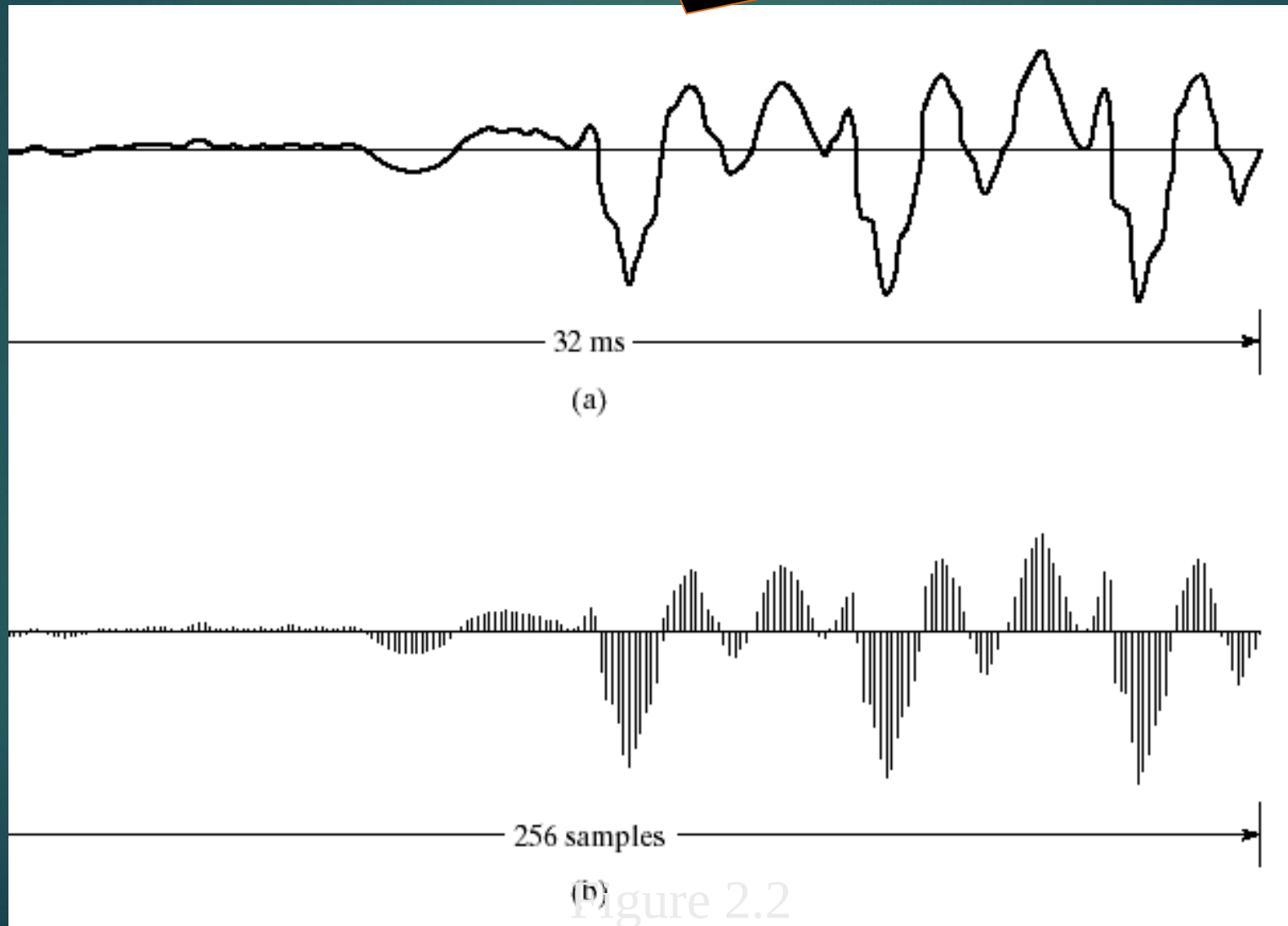
Abcissa: continuous line

$x[n]$: is defined only at discrete instants

$$x[n] = x_a(t) \big|_{t=nT} = x_a(nT)$$

EXAMPLE

Sampling the analog waveform



Basic Sequence Operations

- ▶ Sum of two sequences

$$x[n] + y[n]$$

- ▶ Product of two sequences

$$x[n] \cdot y[n]$$

- ▶ Multiplication of a sequence by a number α

$$\alpha \cdot x[n]$$

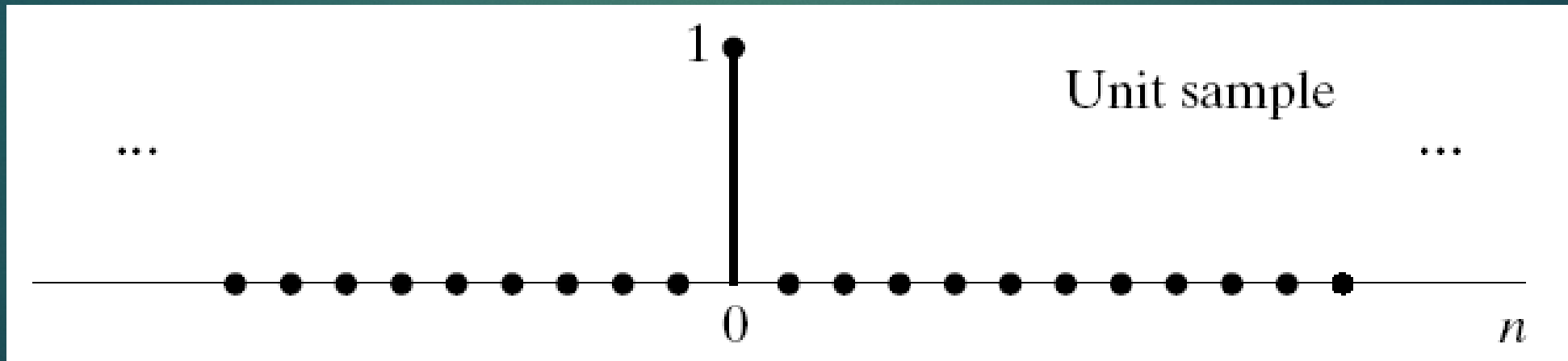
- ▶ Delay (shift) of a sequence

$$y[n] = x[n - n_0] \quad n_0 : \text{integer}$$

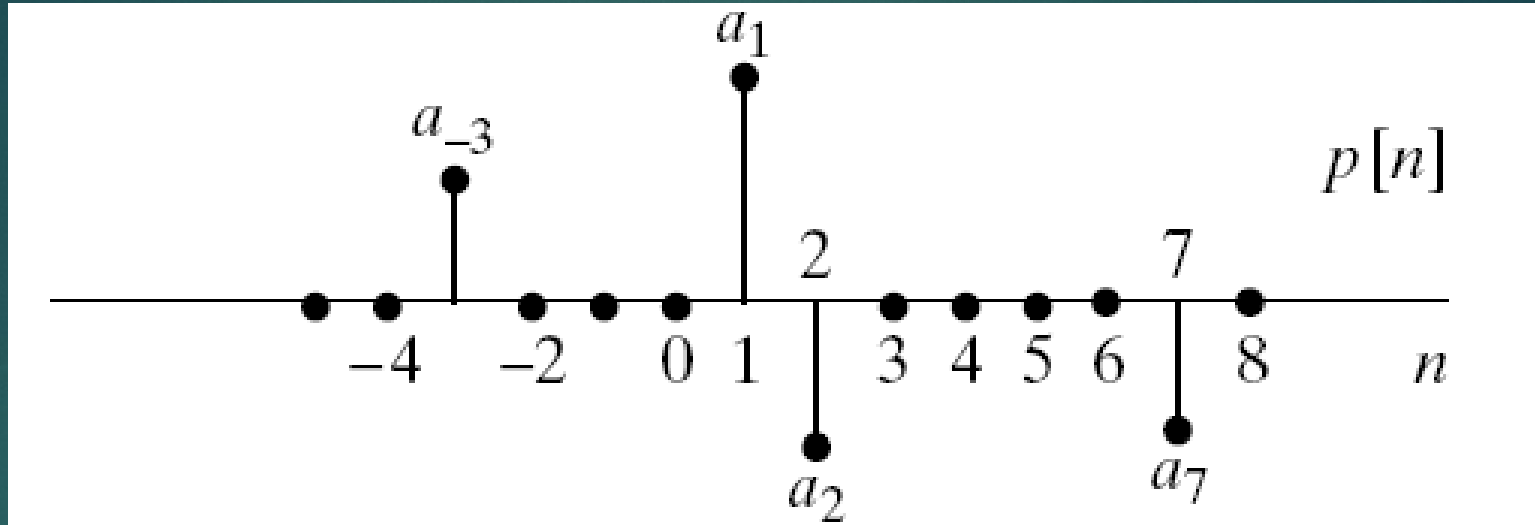
Basic sequences

- ▶ Unit sample sequence
(discrete-time impulse,
impulse)

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$$



Basic sequences



A sum of scaled, delayed impulses

$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_7\delta[n-7]$$

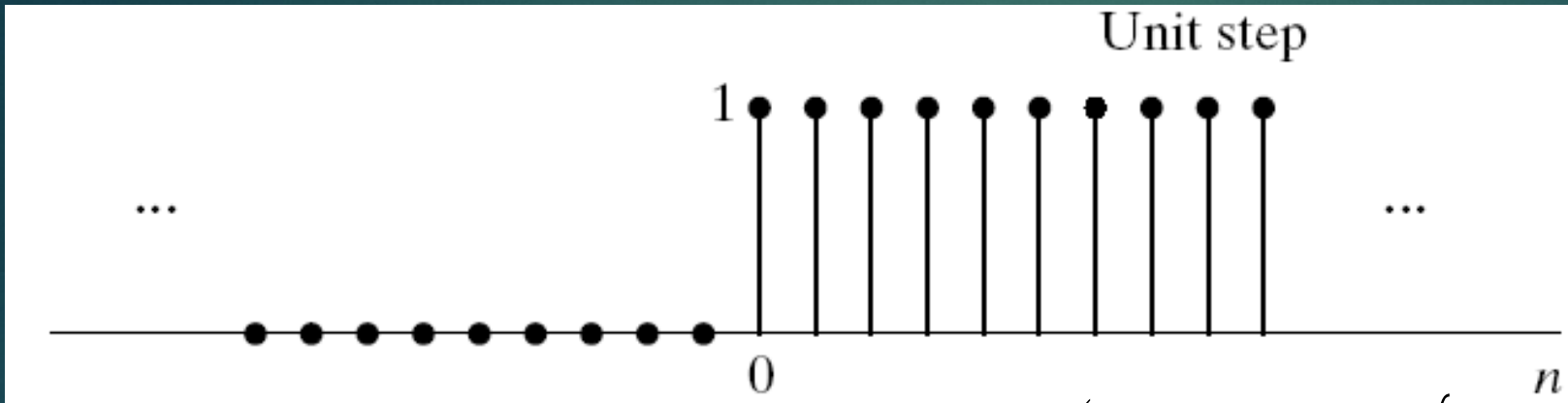
◆ arbitrary
sequence

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Basic sequences

► Unit step sequence

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

$$\left(\begin{aligned} \sum_{k=-\infty}^n \delta[k] &= \begin{cases} 0, & \text{when } n < 0 \\ 1, & \text{when } n \geq 0 \end{cases}, \\ \text{since } \delta[k] &= \begin{cases} 0 & k \neq 0 \\ 1 & k = 0 \end{cases} \end{aligned} \right)$$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots = \sum_{k=0}^{\infty} \delta[n-k]$$

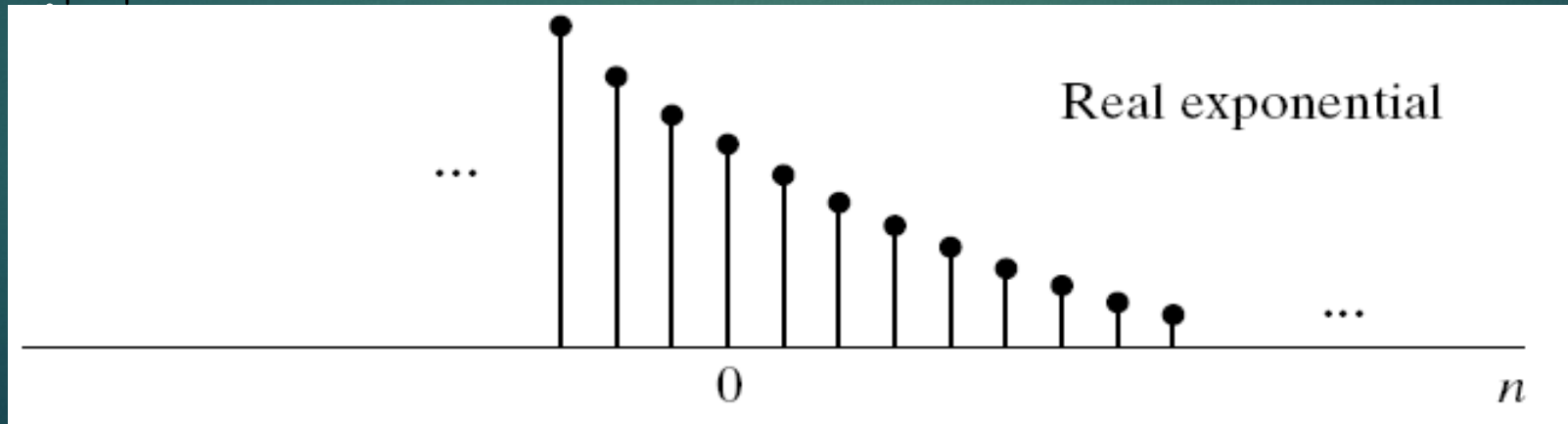
$$\delta[n] = u[n] - u[n-1]$$

First backward difference

Basic Sequences

► Exponential sequences $x[n] = A\alpha^n$

- ◆ A and α are real: $x[n]$ is real
- ◆ A is positive and $0 < \alpha < 1$, $x[n]$ is positive and decrease with increasing n
- ◆ $-1 < \alpha < 0$, $x[n]$ alternate in sign, but decrease in magnitude with increasing n
- ◆ $|\alpha| > 1$ $x[n]$ grows in magnitude as n



Combining Basic sequences

- ◆ If we want an exponential sequences that is zero for $n < 0$, then

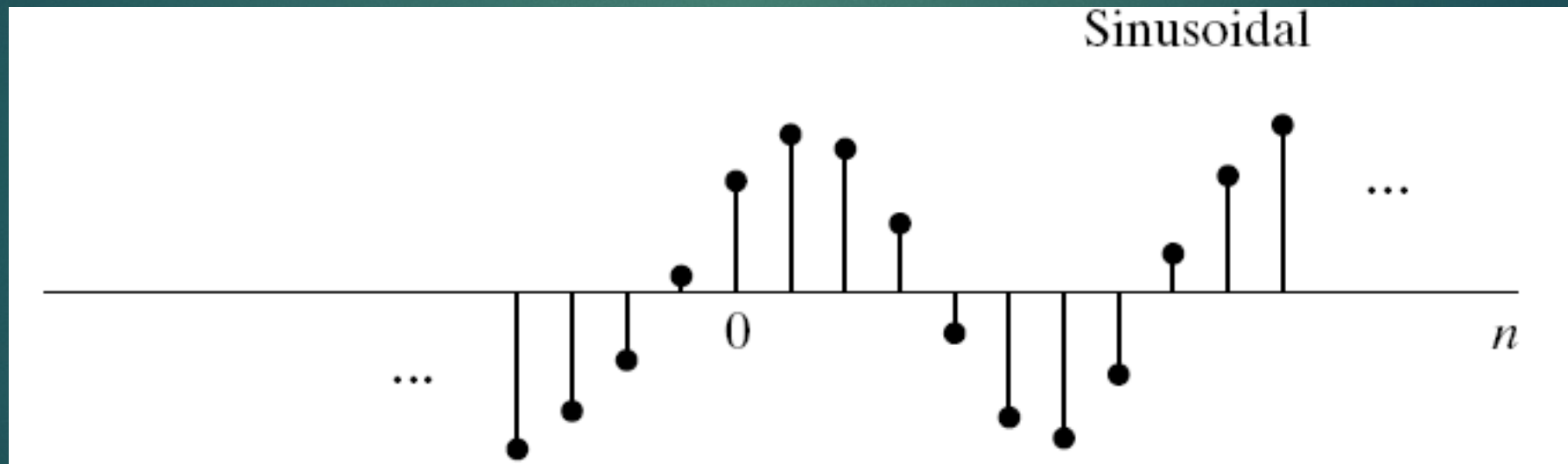
$$x[n] = \begin{cases} A\alpha^n & n \geq 0 \\ 0 & n < 0 \end{cases} \quad \begin{matrix} \text{Cumbersome} \\ e \end{matrix}$$

$$x[n] = A\alpha^n u[n] \quad \text{simpler}$$

Basic sequences

► Sinusoidal sequence

$$x[n] = A \cos(w_0 n + \phi) \quad \text{for all } n$$



Exponential Sequences

$$A = |A|e^{j\phi} \quad \alpha = |\alpha|e^{jw_0}$$

$$x[n] = A\alpha^n = |A|e^{j\phi} |\alpha|^n e^{jw_0 n} = |A||\alpha|^n e^{j(w_0 n + \phi)}$$

$$= |A||\alpha|^n \cos(w_0 n + \phi) + j|A||\alpha|^n \sin(w_0 n + \phi)$$

Exponentially weighted

sinusoids

Exponentially growing

envelope

Exponentially decreasing

envelope

$$|\alpha| > 1$$

$$|\alpha| < 1$$

$$|\alpha| = 1$$

$$x[n] = Ae^{jw_0 n}$$

is referred to

Complex Exponential Sequences

Frequency of complex sinusoids

$$x[n] = Ae^{j(w_0 + 2\pi)n} = Ae^{jw_0n} e^{j2\pi n} = Ae^{jw_0n}$$

$$x[n] = A \cos \left[(w_0 + 2\pi r)n + \varphi \right] = A \cos (w_0n + \varphi)$$

- ▶ w_0 : frequency of the complex sinusoid or complex exponential
- ▶ ϕ : phase

Periodic Sequences

- ▶ A periodic sequence with integer period N

$$x[n] = x[n + N] \quad \text{for all } n$$

$$A \cos(w_0 n + \phi) = A \cos(w_0 n + w_0 N + \phi)$$

$$w_0 N = 2\pi k, \text{ where } k \text{ is integer}$$

$$N = 2\pi k / w_0, \text{ where } k \text{ is integer}$$

Examples of Periodic Sequences

$$x_1[n] = \cos(\pi n / 4)$$

- Suppose it is periodic sequence with period N

$$x_1[n] = x_1[n + N]$$

$$\cos(\pi n / 4) = \cos[\pi(n + N) / 4]$$

$$\pi n / 4 + 2\pi k = \pi n / 4 + N\pi / 4, \quad k : \text{integer}$$

$$N = 2\pi k / (\pi / 4) = 8k$$

$$k = 1, \rightarrow N = 8 = 2\pi / w_0$$

Examples of Periodic Sequences

$$\frac{2\pi}{8} \rightarrow \frac{3\pi}{8} \rightarrow x_1[n] = \cos(3\pi n / 8)$$

- Suppose it is periodic sequence with period N

$$x_1[n] = x_1[n + N]$$

$$\cos(3\pi n / 8) = \cos[3\pi(n + N) / 8]$$

$$3\pi n / 8 + 2\pi k = 3\pi n / 8 + 3N\pi / 8, \quad k : \text{integer}$$

$$N = 2\pi k / w_0 = 2\pi k / (3\pi / 8) \quad k = 3, \rightarrow N = 16$$

$$N = 2\pi 3 / w_0 \neq 2\pi / w_0 \quad (\text{for continuous signal})$$

Non-Periodic Sequences

$$x_2[n] = \cos(n)$$

- Suppose it is periodic sequence with period N

$$x_2[n] = x_2[n + N]$$

$$\cos(n) = \cos(n + N)$$

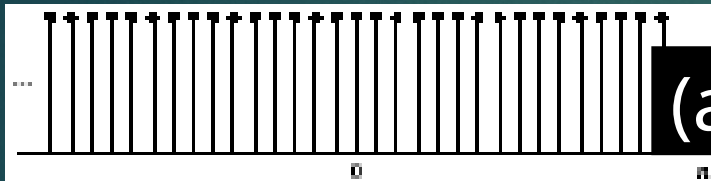
for $n + 2\pi k = n + N, \quad k : \text{integer},$

there is no integer N

High and Low Frequencies in Discrete-time signal

$$x[n] = A \cos(w_0 n)$$

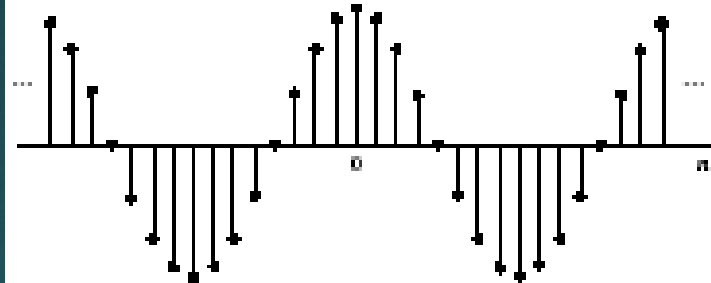
(a) $w_0 = 0$ or 2π



(a)

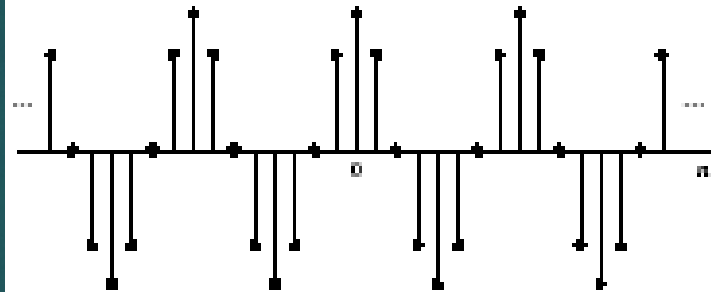
Frequency: The rate at which a repeating event occurs.

(b) $w_0 = \pi/8$ or $15\pi/8$



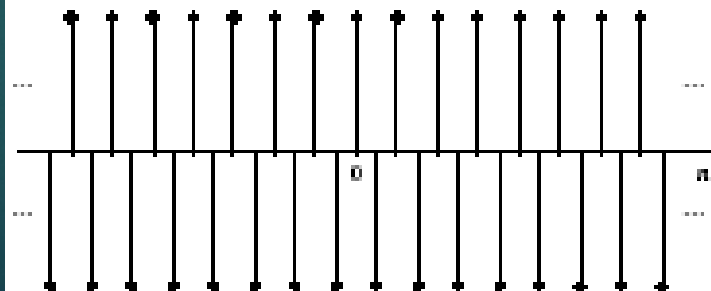
(b)

(c) $w_0 = \pi/4$ or $7\pi/4$



(c)

(d) $w_0 = \pi$



(d)

System / Filter

Definition

System A system (or a process) is a set of units that process the input signal to produce an output.

- A subset of environment (already in existence) that is under study.
- A filter is that unit which has been set up to process an input signal to produce a (desired) output.
 - ▶ Usually designed (as an algorithm or a hardware) to perform a pre-specified task such as data cleaning, signal removal, etc.
- A system is a filter and a filter is also a system!
 - ▶ Theoretically, there is no distinction in their operation on the input signals. Understanding of systems help design of filters and vice versa!

Discrete-Time System

- ▶ **Discrete-Time System** is a transformation or operator that maps input sequence $x[n]$ into a unique $y[n]$
- ▶ $y[n]=T\{x[n]\}$, $x[n]$, $y[n]$: discrete-time signal



Discrete-Time System

The Ideal Delay System

$$y[n] = x[n - n_d], \quad -\infty < n < \infty$$

- ◆ If n_d is a **positive** integer: the **delay** of the system, **Shift** the input sequence to the **right** by n_d samples to form the output.
- ◆ If n_d is a **negative** integer: the system will **shift** the input to the **left** by $|n_d|$ samples, corresponding to a **time advance**.

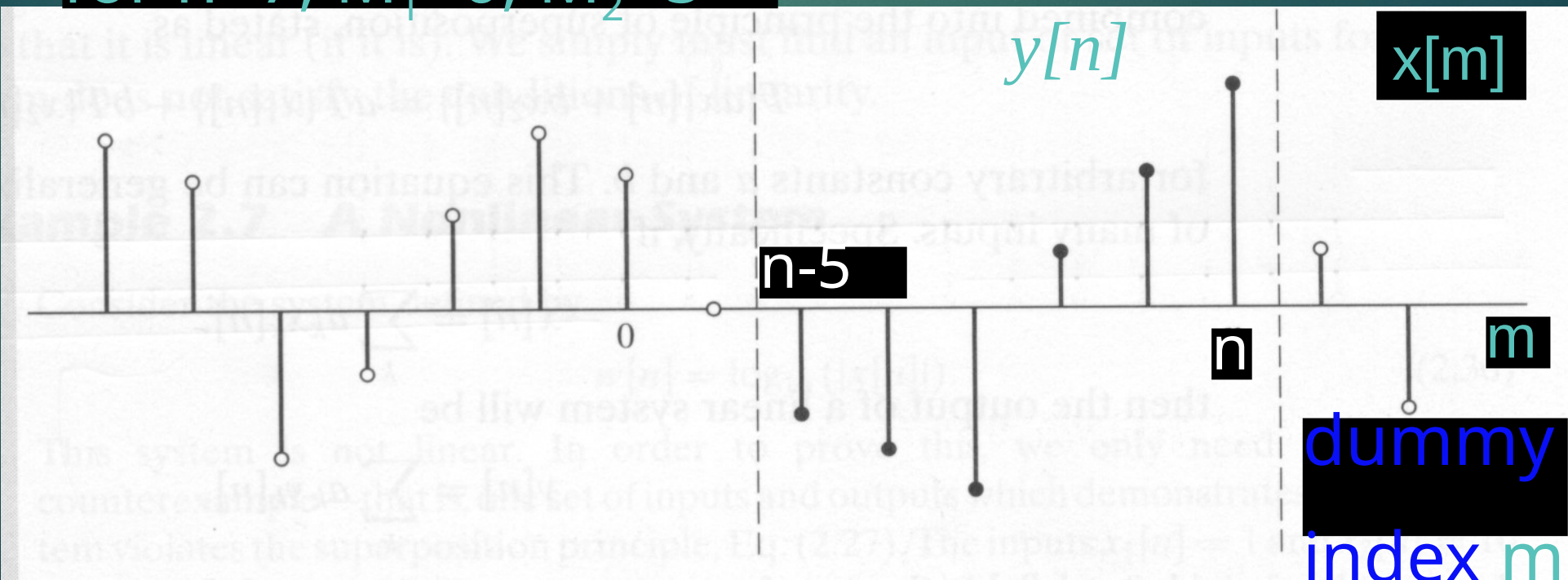
Moving Average

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n-k]$$

$$y[n] = \frac{1}{S} \sum_{i=A}^{nS+1} x[n]$$

$$= \frac{1}{M_1 + M_2 + 1} \left\{ x[n+M_1] + x[n+M_1-1] + \dots + x[n] + x[n-1] + \dots + x[n-M_2] \right\}$$

for $n=7, M_1=0, M_2=5$



Properties of Discrete-time systems

Memoryless (memory) system

► Memoryless systems:

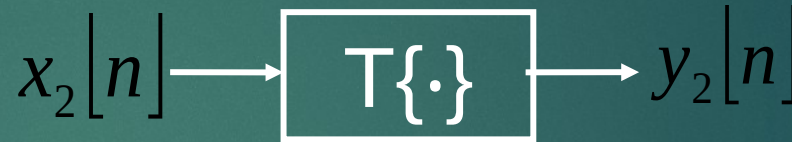
the output $y[n]$ at every value of n depends only on the input $x[n]$ at the same value of n

$$y[n] = (x[n])^2$$

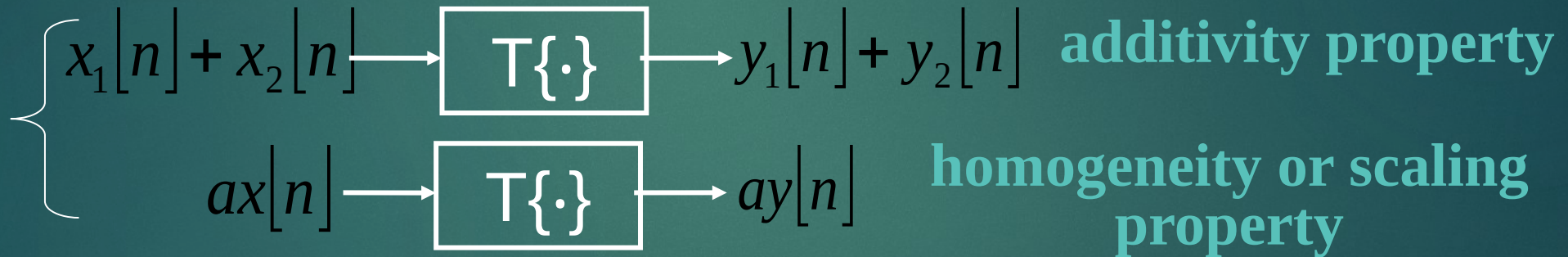
Properties of Discrete-time systems

Linear Systems

◆ If



◆ and only If:



◆ principle of superposition



Example of Linear System

► Accumulator system

for arbitrary $x_1[n]$ and $x_2[n]$

$$y[n] = \sum_{k=-\infty}^n x[k]$$

$$y_1[n] = \sum_{k=-\infty}^n x_1[k]$$

$$y_2[n] = \sum_{k=-\infty}^n x_2[k]$$

when $x_3[n] = ax_1[n] + bx_2[n]$

$$y_3[n] = \sum_{k=-\infty}^n x_3[k] = \sum_{k=-\infty}^n (ax_1[k] + bx_2[k])$$

$$= a \sum_{k=-\infty}^n x_1[k] + b \sum_{k=-\infty}^n x_2[k] = ay_1[n] + by_2[n]$$

Nonlinear Systems

◆ Method: find one counterexample

◆ **For** $y[n] = (x[n])^2$

◆ counterexample $1^2 + 1^2 \neq (1+1)^2$

◆ **For** $y[n] = \log_{10}(|x[n]|)$

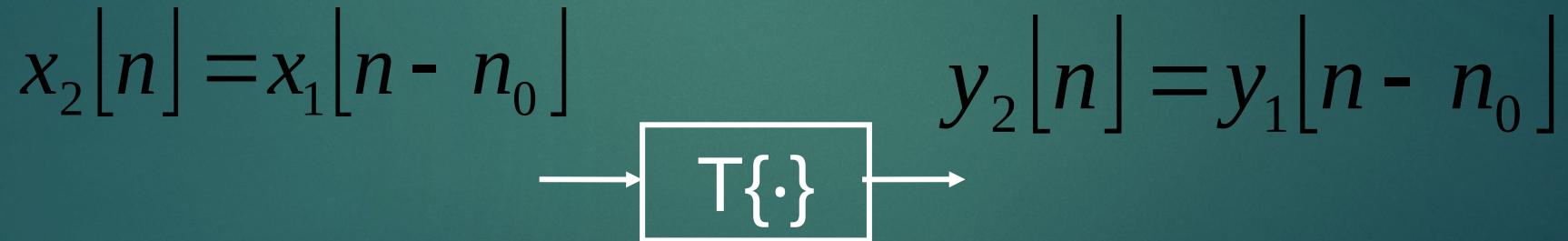
◆ counterexample

$$10 \times \log_{10}(|1|) \neq \log_{10}(|10 \times 1|)$$

Properties of Discrete-time systems

Time-Invariant Systems

► Shift-Invariant Systems



Example of Time-Invariant System

► Accumulator system

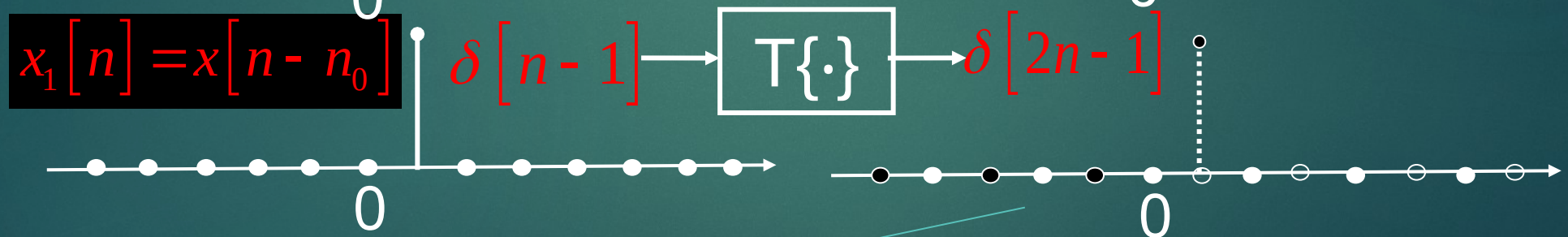
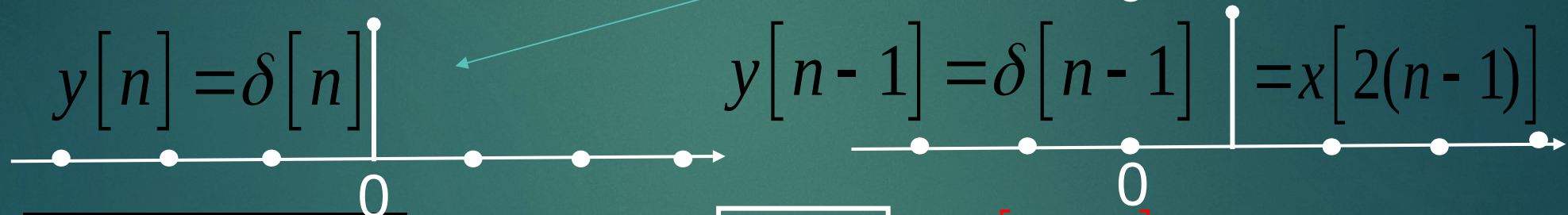
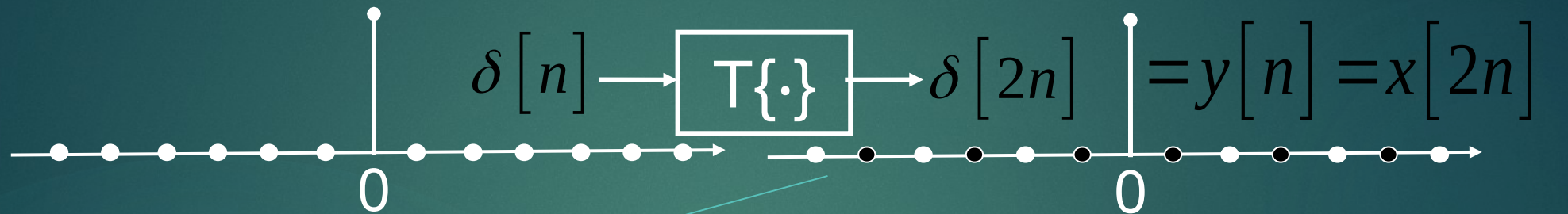
$$y[n] = \sum_{k=-\infty}^n x[k]$$

$$x_1 = x[n - n_0]$$

$$y_1[n] = \sum_{k=-\infty}^n x_1[k] = \sum_{k=-\infty}^n x[k - n_0] = \sum_{k_1=-\infty}^{n-n_0} x[k_1] = y[n - n_0]$$

The compressor system

$$x[n] \rightarrow \boxed{T\{\cdot\}} \rightarrow y[n] = x[Mn], \quad -\infty < n < \infty$$



$$y_1[n] = x_1[Mn] = x[Mn - n_0] \neq y[n - n_0] = x[M(n - n_0)]$$

Properties of Discrete-time systems

Causality

- ▶ A system is causal if, for every choice of n_0 the output sequence value at the index $n = n_0$ depends only on the input sequence value for $n \leq n_0$

Example for Causal System

- ▶ Forward difference system is not Causal

$$y[n] = x[n+1] - x[n]$$

- ▶ Backward difference system is Causal

$$y[n] = x[n] - x[n-1]$$

Properties of Discrete-time systems

Stability

- ▶ Bounded-Input Bounded-Output (**BIBO**)
Stability: every bounded input sequence produces a bounded output sequence.

$$\text{if } |x[n]| \leq B_x < \infty, \quad \text{for all } n$$

$$\text{then } |y[n]| \leq B_y < \infty, \quad \text{for all } n$$

Testing for Stability or Instability

$$y[n] = (x[n])^2 \quad \text{is stable}$$

i
f

$$|x[n]| \leq B_x < \infty, \quad \text{for all } n$$

then
n

$$|y[n]| \leq B_y = B_x^2 < \infty, \quad \text{for all } n$$

Testing for Stability or Instability

► Accumulator system $y[n] = \sum_{k=-\infty}^n x[k]$

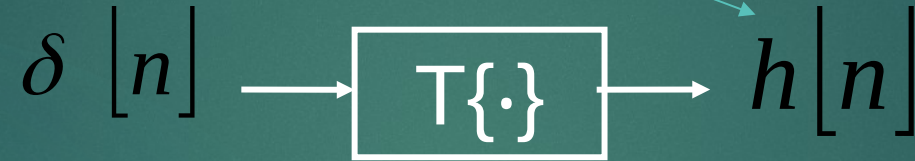
$$x[n] = u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} : \text{bounded}$$

$$y[n] = \sum_{k=-\infty}^n x[k] = \sum_{k=-\infty}^n x[k] = \begin{cases} 0 & n < 0 \\ n+1 & n \geq 0 \end{cases} : \text{not bounded}$$

◆ Accumulator system **is not stable**

Linear Time-Invariant (LTI) Systems

► Impulse response



LTI Systems: Convolution

- ◆ Representation of general sequence as a linear combination of delayed impulse

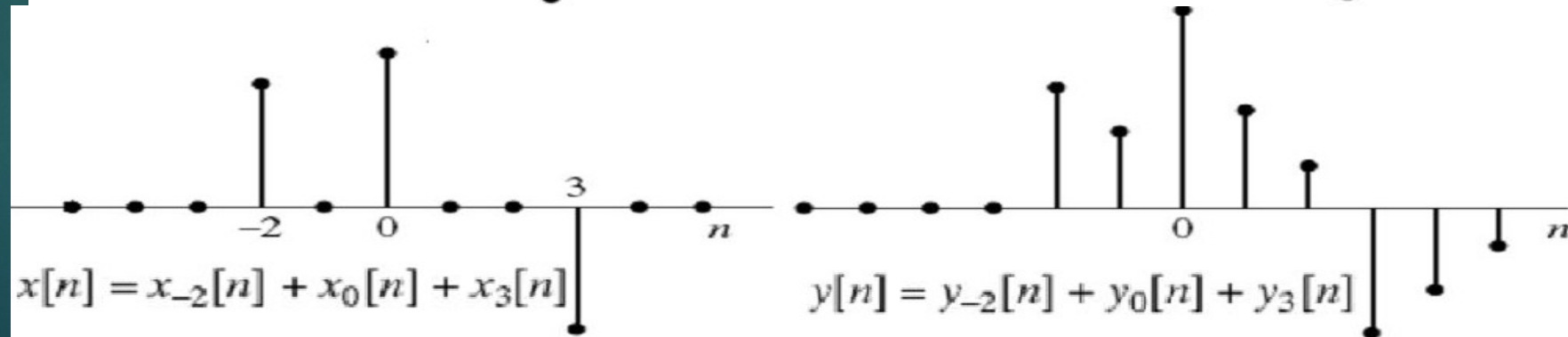
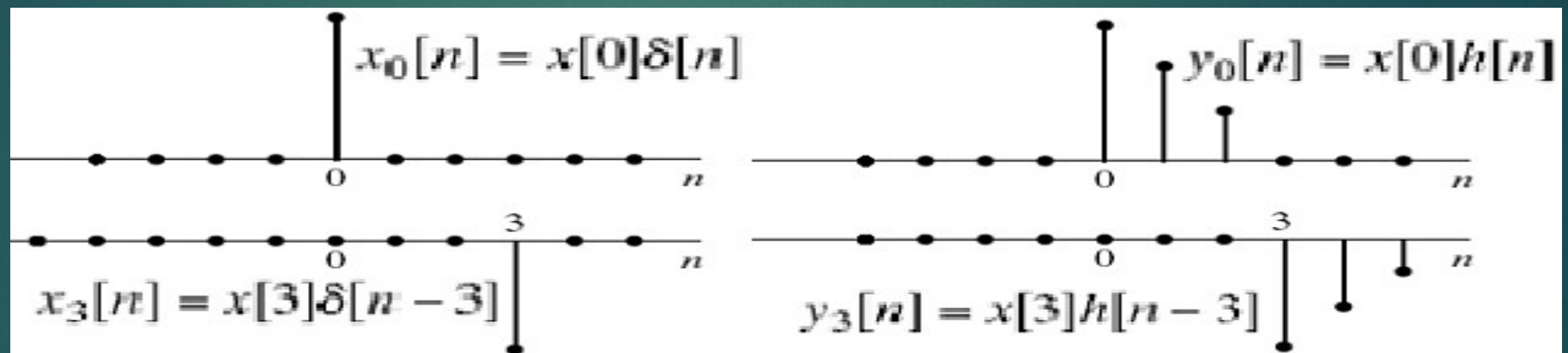
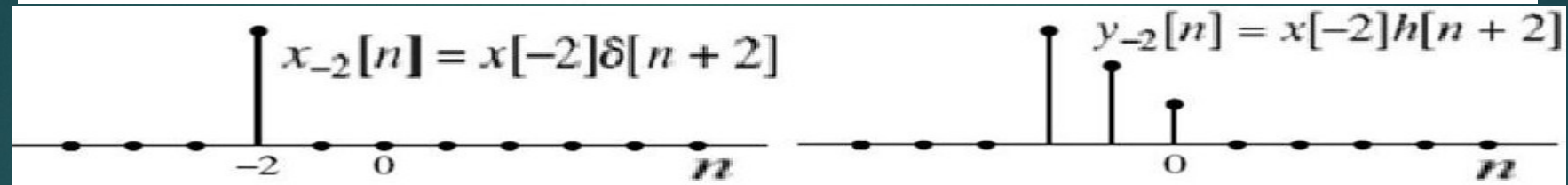
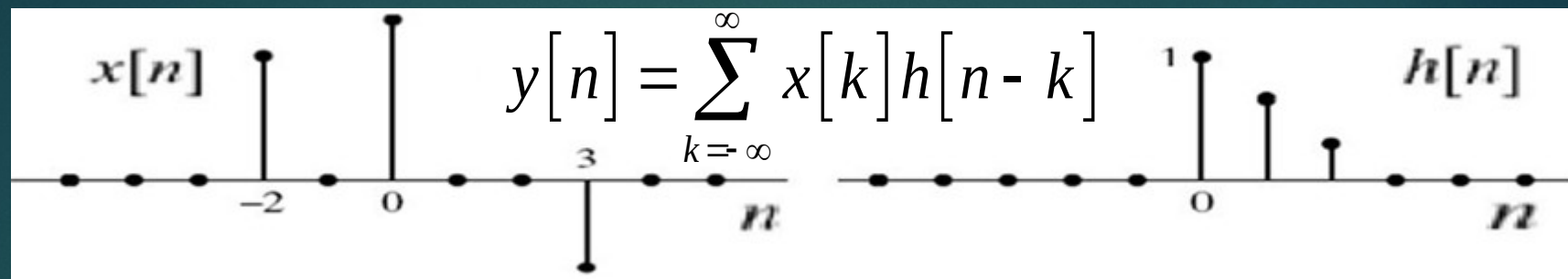
$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad \delta[n] \xrightarrow{\text{red arrow}} h[n]$$

- ◆ principle of superposition

$$\delta[n-k] \xrightarrow{\text{red arrow}} h[n-k]$$

$$\begin{aligned} y[n] &= T \left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \right\} = \sum_{k=-\infty}^{\infty} x[k] T \left\{ \delta[n-k] \right\} \\ &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] \star h[n] \end{aligned}$$

An Illustration Example (interpretation 1) 



Computation of the Convolution

(interpretation

$$y[n] \stackrel{2)}{=} \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

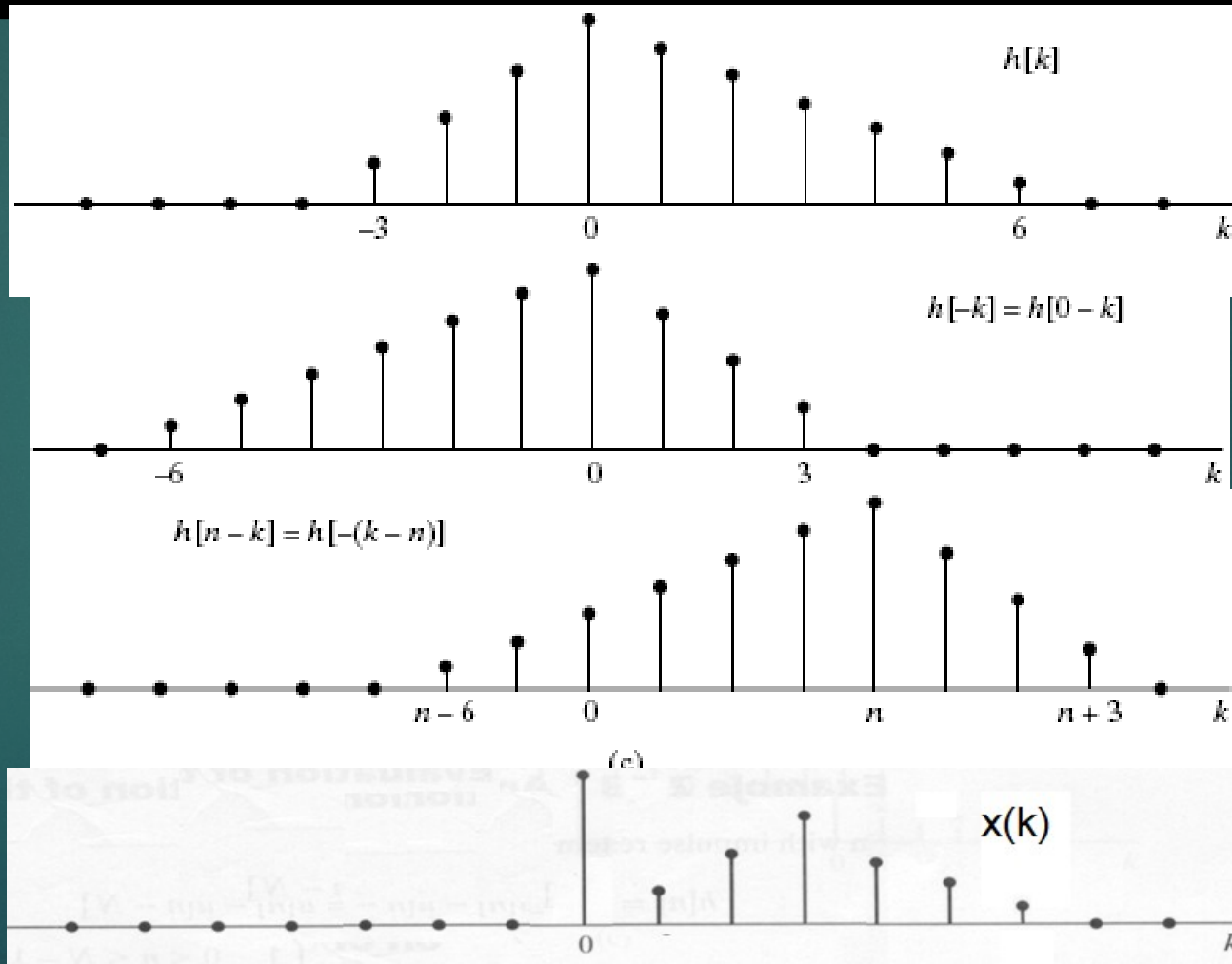
$$h[n-k] = h[-(k-n)]$$

$$h[k] \xrightarrow{\text{red arrow}} h[-k] \xrightarrow{\text{blue arrow}}$$

- ▶ reflecting $h[k]$ about the origin to obtain $h[-k]$
- ▶ Shifting the origin of the reflected sequence to $k=n$

Convolution can be realized by

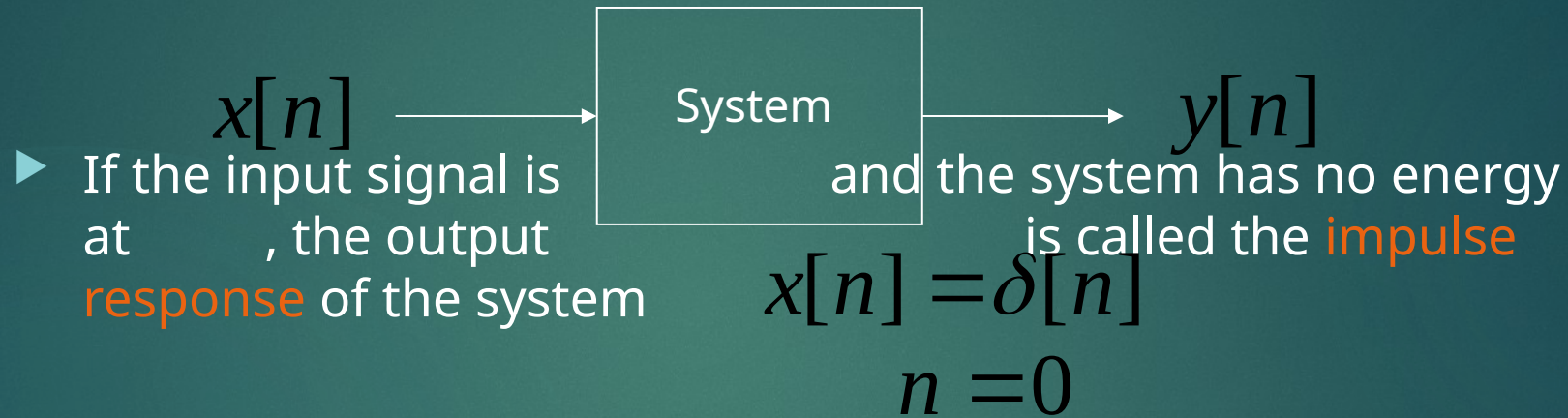
- Reflecting(reversing) $h[k]$ about the origin to obtain $h[-k]$.
- Shifting the origin of the reflected sequences to $k=n$.
- Computing the weighted moving average of $x[k]$ by using the weights given by $h[n-k]$.



$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

DT Unit-Impulse Response

- ▶ Consider the DT SISO system:



$$y[n] = h[n]$$



Representing Signals in Terms of Shifted and Scaled Impulses

- ▶ Let $x[n]$ be an arbitrary input signal to a DT LTI system
- ▶ Suppose that $x[n] = 0$ for $n = -1, -2, \dots$
- ▶ This signal can be represented as

$$x[n] = 0 \quad n = -1, -2, \dots$$

$$x[n] = x[0]\delta[n] + x[1]\delta[n - 1] + x[2]\delta[n - 2] + \dots$$

$$= \sum_{i=0}^{\infty} x[i]\delta[n - i], \quad n = 0, 1, 2, \dots$$

Exploiting Time-Invariance and Linearity

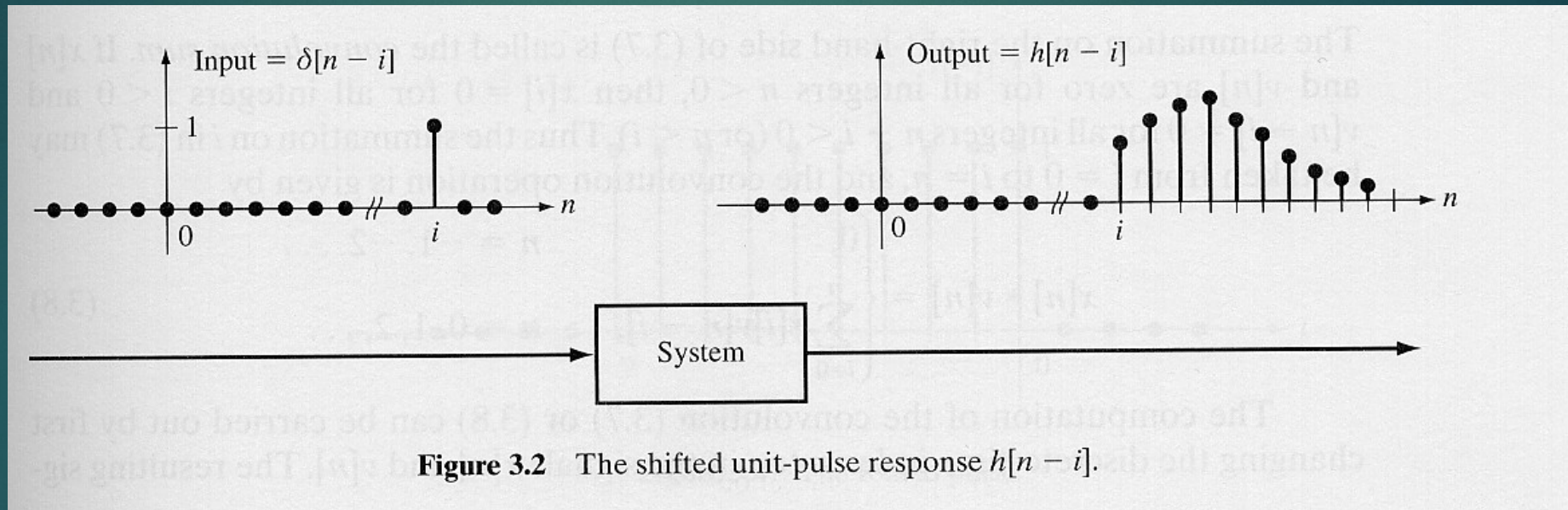


Figure 3.2 The shifted unit-pulse response $h[n - i]$.

$$y[n] = \sum_{i=0}^{\infty} x[i]h[n - i], \quad n \geq 0$$

The Convolution Sum

- ▶ This particular summation is called the **convolution sum**

$$y[n] = \sum_{i=-\infty}^{\infty} x[i]h[n-i]$$

- ▶ Equation $y[n] = \sum_{i=-\infty}^{\infty} x[i]h[n-i]$ is called the *convolution representation of the system*

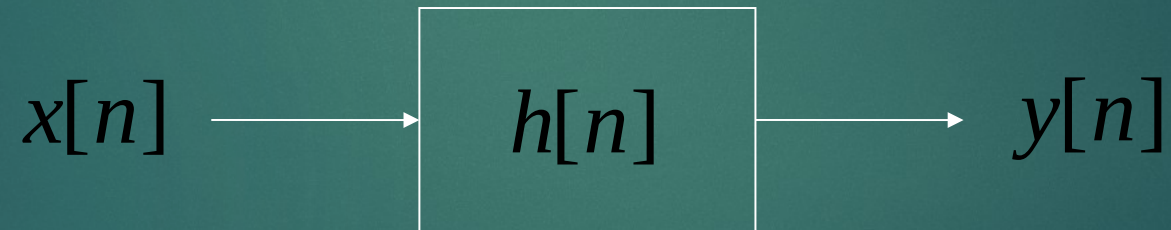
$$x[n] \star h[n]$$

- ▶ Remark: a DT LTI system is completely described by its impulse response $h[n]$

$$y[n] = x[n] \star h[n]$$

Block Diagram Representation of DT LTI Systems

- ▶ Since the impulse response $h[n]$ provides the complete description of a DT LTI system, we write



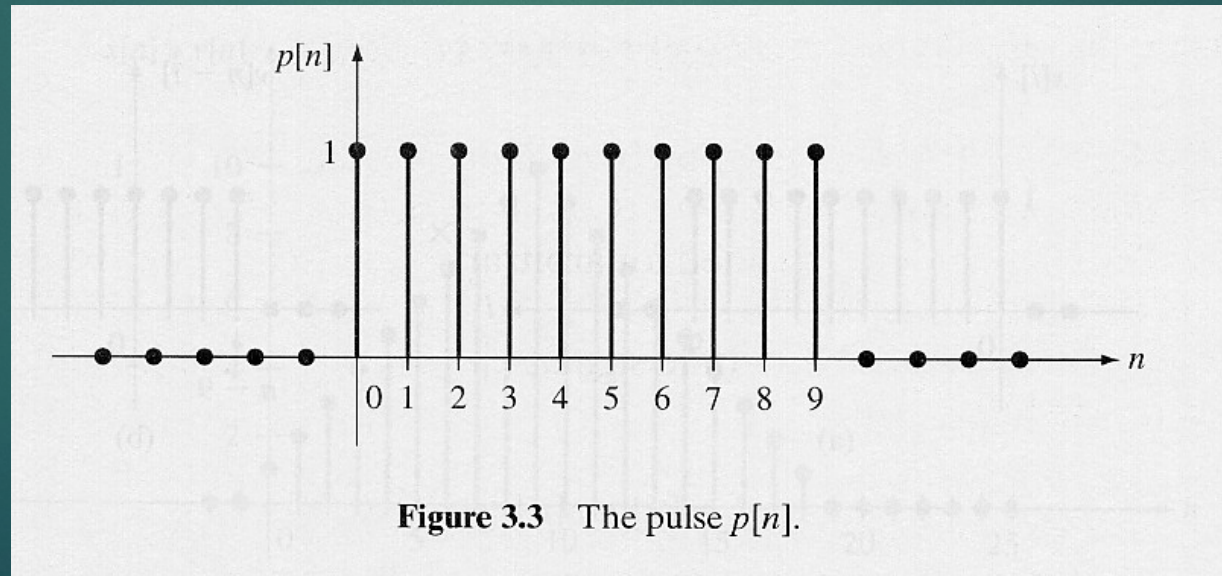
The Convolution Sum for Noncausal Signals

- ▶ Suppose that we have two signals $x[n]$ and $v[n]$ that are not zero for negative times (**noncausal signals**)
- ▶ Then, their convolution is expressed by the two-sided series

$$y[n] = \sum_{i=-\infty}^{\infty} x[i]v[n - i]$$

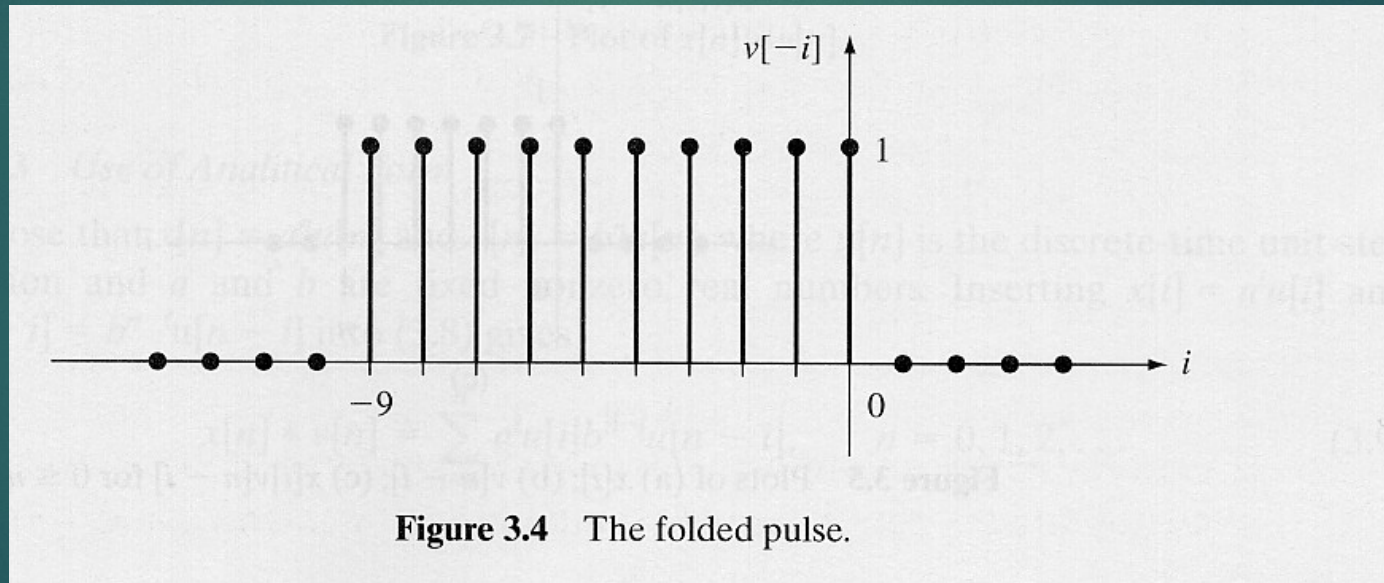
Example: Convolution of Two Rectangular Pulses

- Suppose that both $x[n]$ and $v[n]$ are equal to the rectangular pulse $p[n]$ (causal signal) depicted below



The Folded Pulse

- ▶ The signal $v[-i]$ is equal to the pulse $v[i]$ folded about the vertical axis



Sliding $v[n-i]$ over $x[i]$

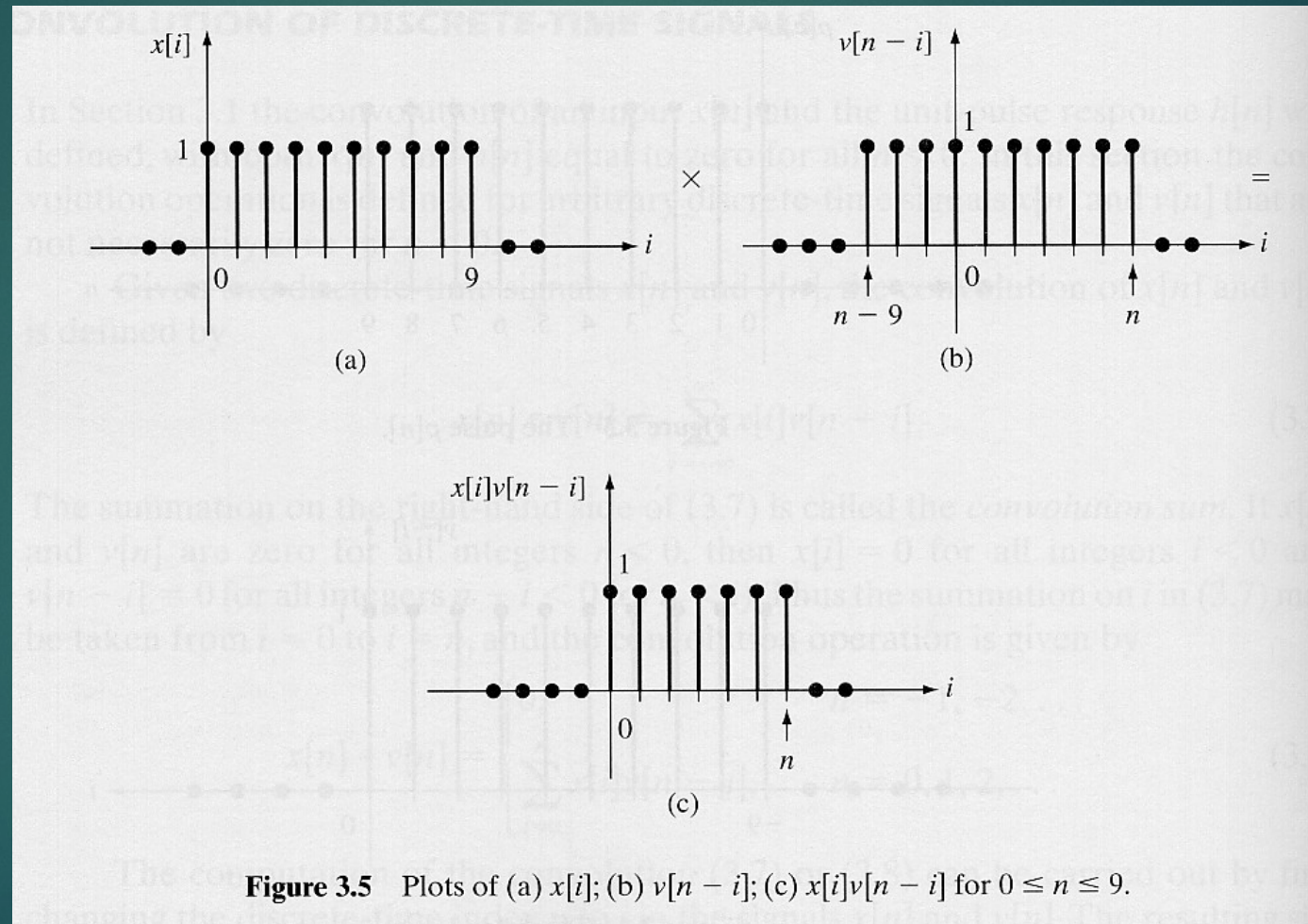


Figure 3.5 Plots of (a) $x[i]$; (b) $v[n-i]$; (c) $x[i]v[n-i]$ for $0 \leq n \leq 9$.

Sliding $v[n-i]$ over $x[i]$ didn't

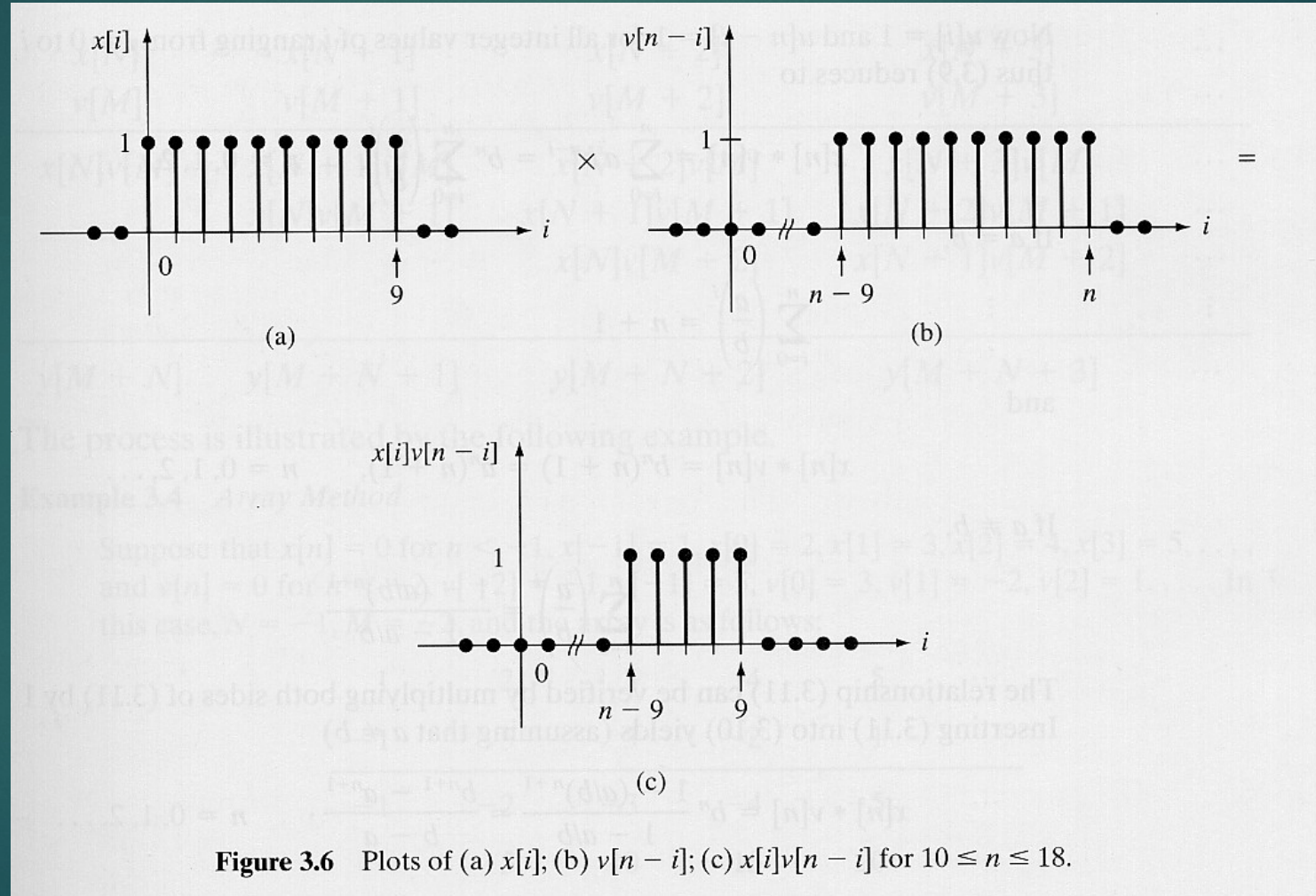


Figure 3.6 Plots of (a) $x[i]$; (b) $v[n-i]$; (c) $x[i]v[n-i]$ for $10 \leq n \leq 18$.

Plot of

$$x[n] * v[n]$$

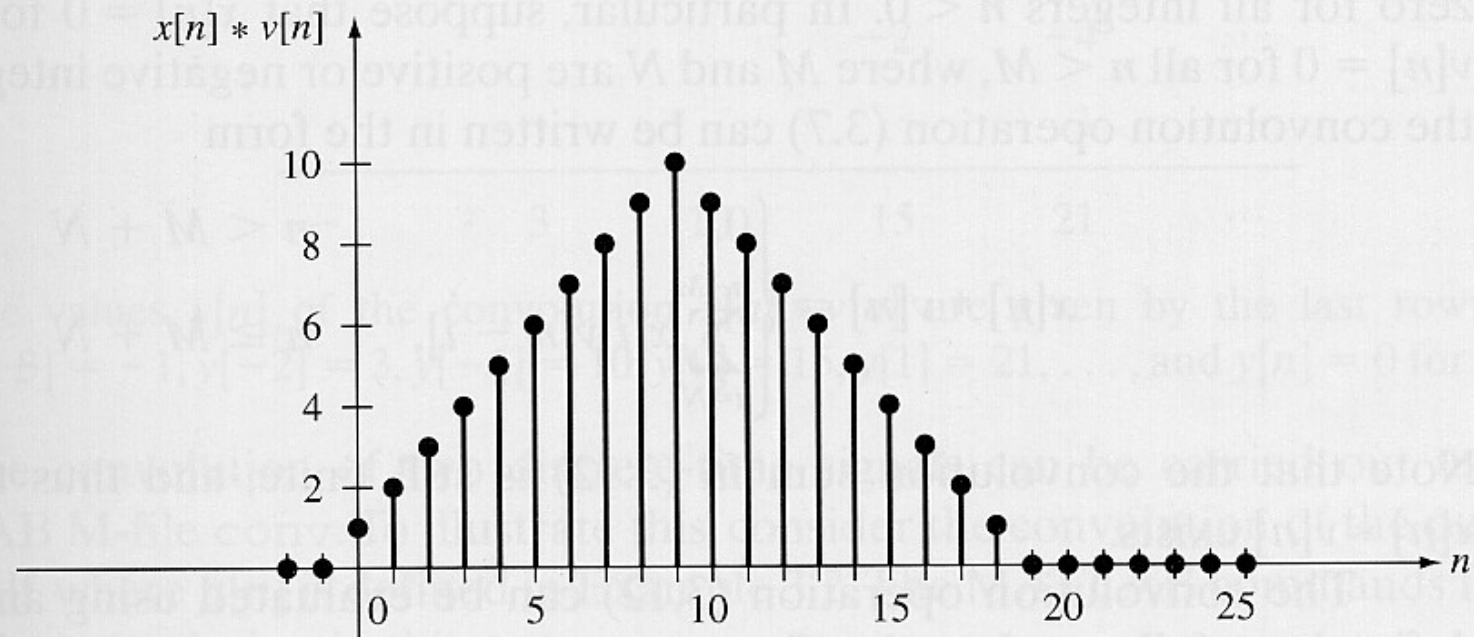


Figure 3.7 Plot of $x[n] * v[n]$.

Three-Steps of Linear Convolution

- ▶ For any given n , how to obtain

$$g(n) = \sum_{k=-\infty}^{\infty} h(k) f(n - k)$$

- ▶ Step 1: time reversal of either signal (e.g., $f(k) \rightarrow f(-k)$)
- ▶ Step 2: shift $f(-k)$ by n samples to obtain $f(n-k)$
- ▶ Step 3: multiply $h(k)$ and $f(n-k)$ for each k and then take the summation over k

Note

You need to change variable n to get the whole sequence.

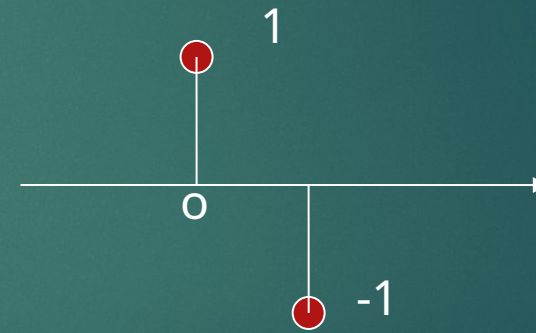
1D Linear Convolution

87

$$f(n)=[1 \ 2 \ 3 \ 4]$$

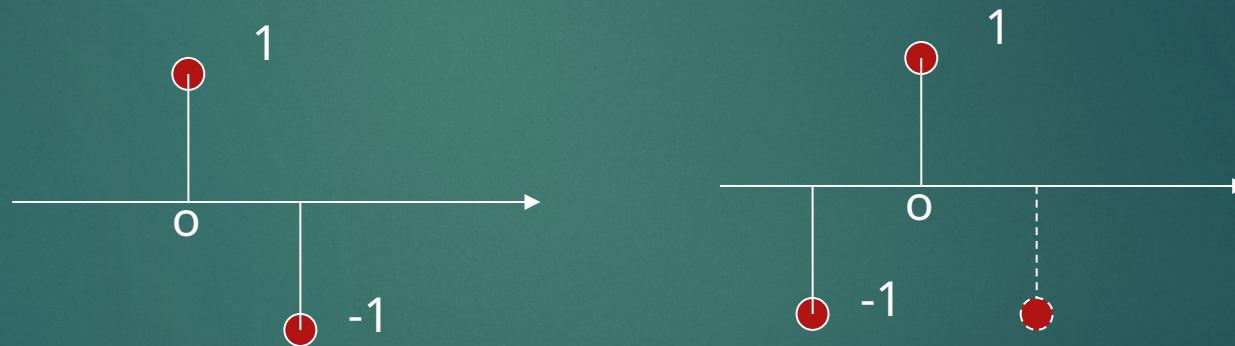


$$h(n)=[1 \ -1]$$

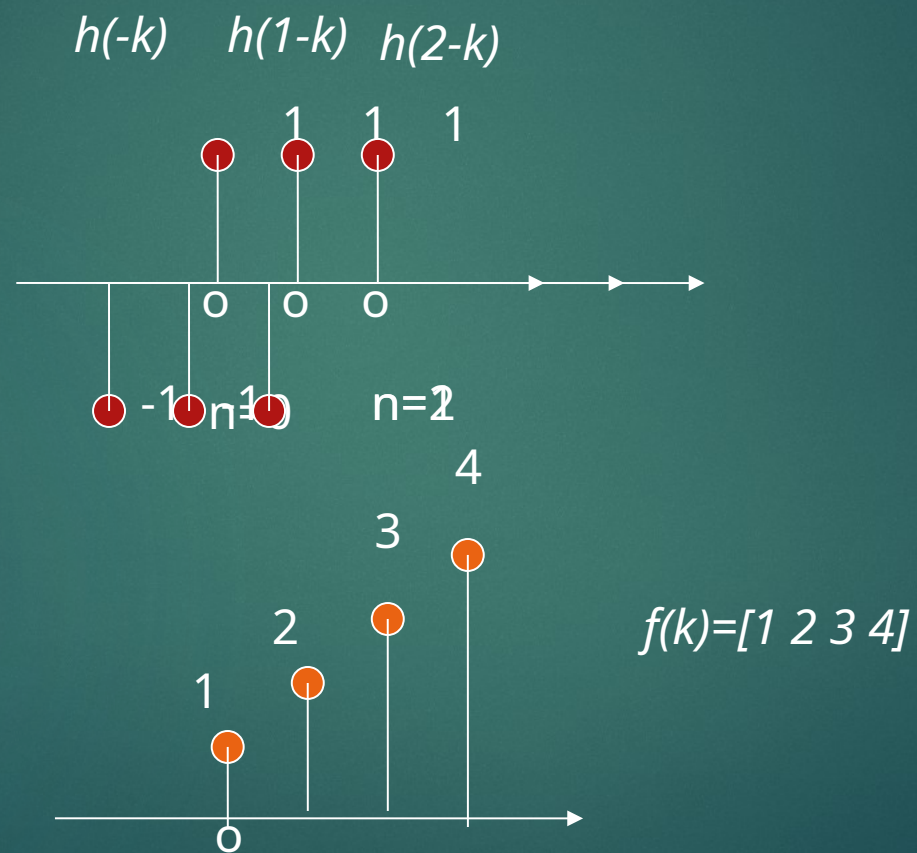


Step 1: Time Reversal

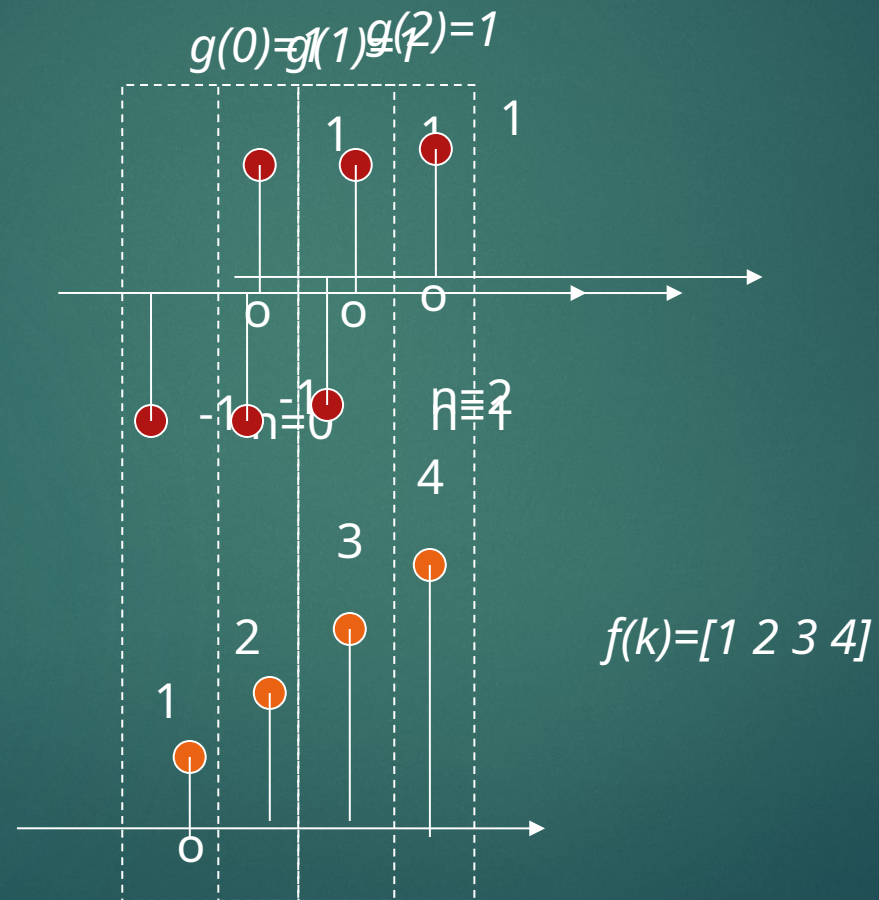
$$h(k)=[1 \ -1] \longrightarrow h(-k)=[-1 \ 1]$$



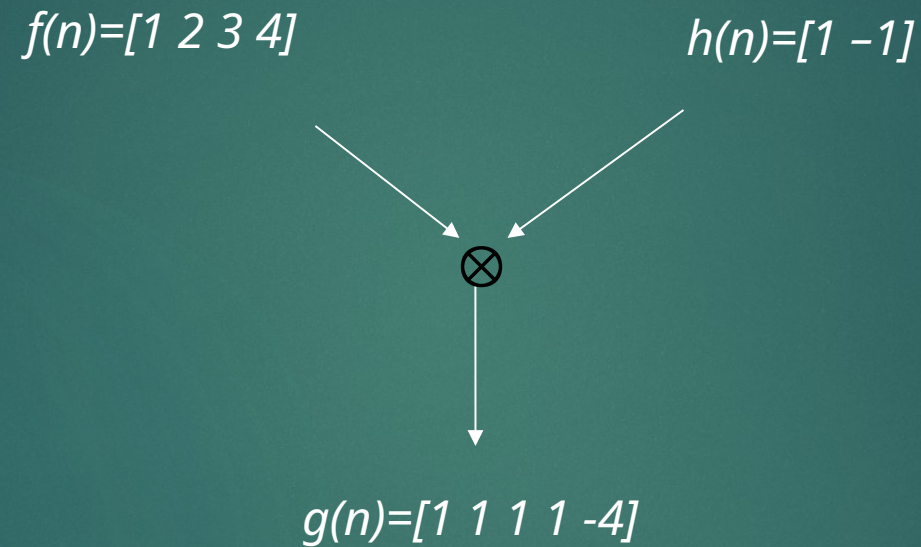
Step 2: Shift



Step3: Multiply-and-Add



Final Result



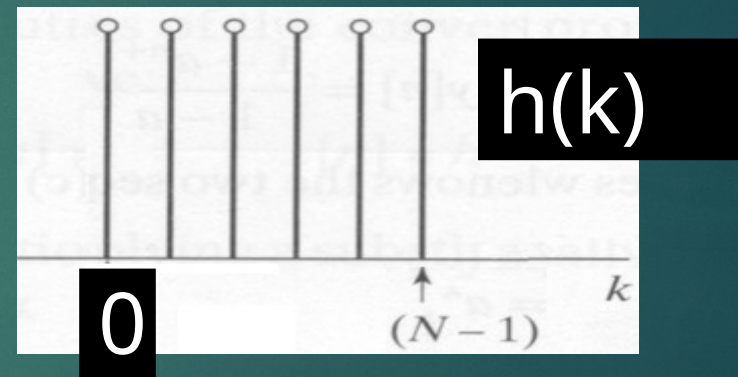
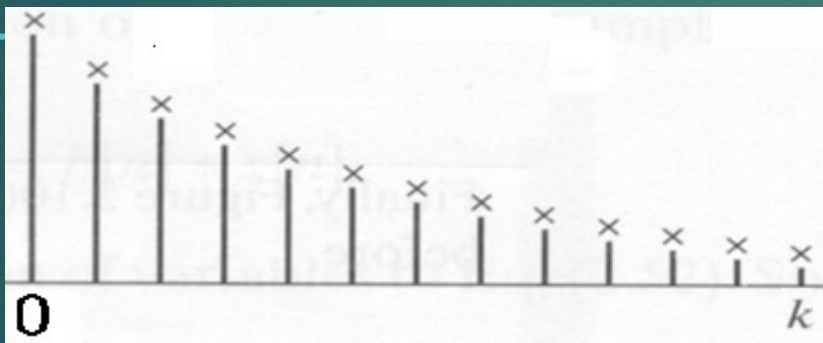
If the lengths of two input signals are N_1 and N_2 respectively, the length of the output signal will be N_1+N_2-1 .

Analytical Evaluation of the Convolution

For system with impulse response

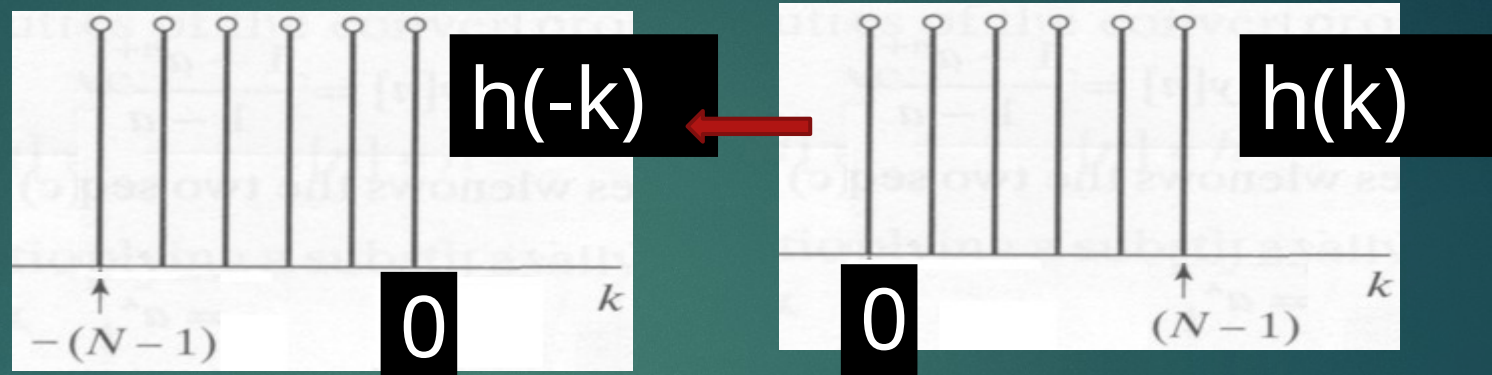
$$h[n] = u[n] - u[n - N] = \begin{cases} 1 & 0 \leq n \leq N - 1 \\ 0 & \text{otherwise} \end{cases}$$

input $x[n] = a^n u[n]$



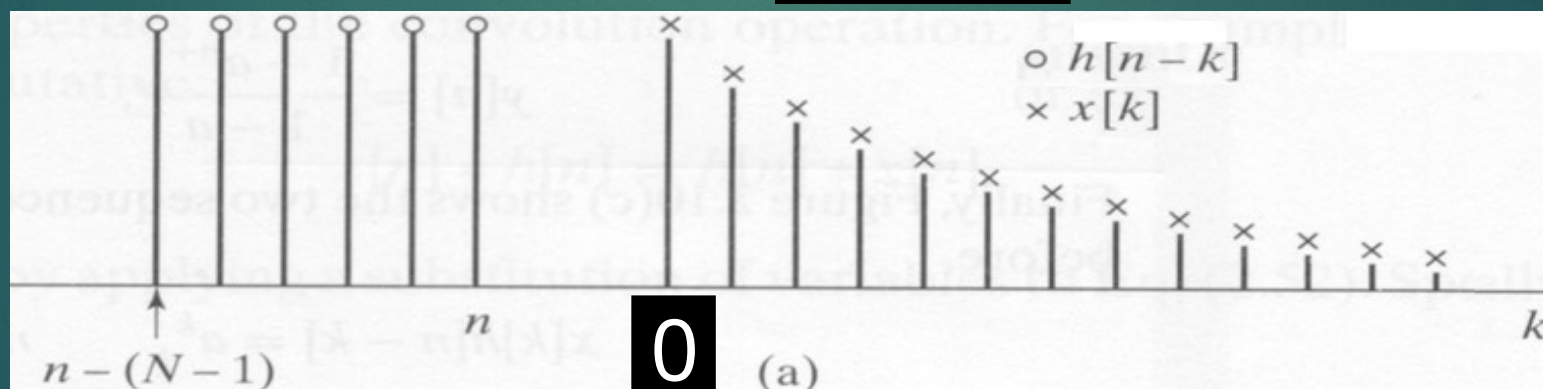
Find the output at index n

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

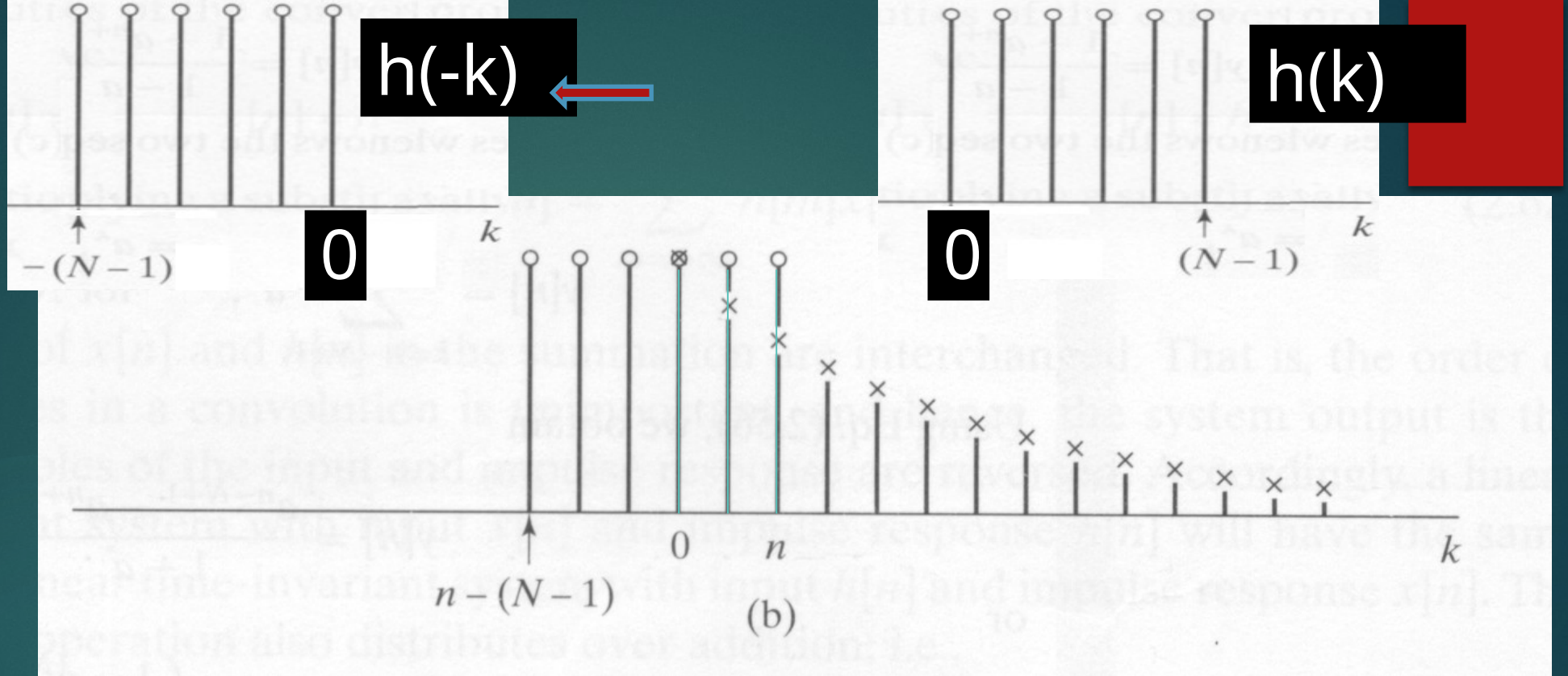


$h(n-k)$

$x(k)$

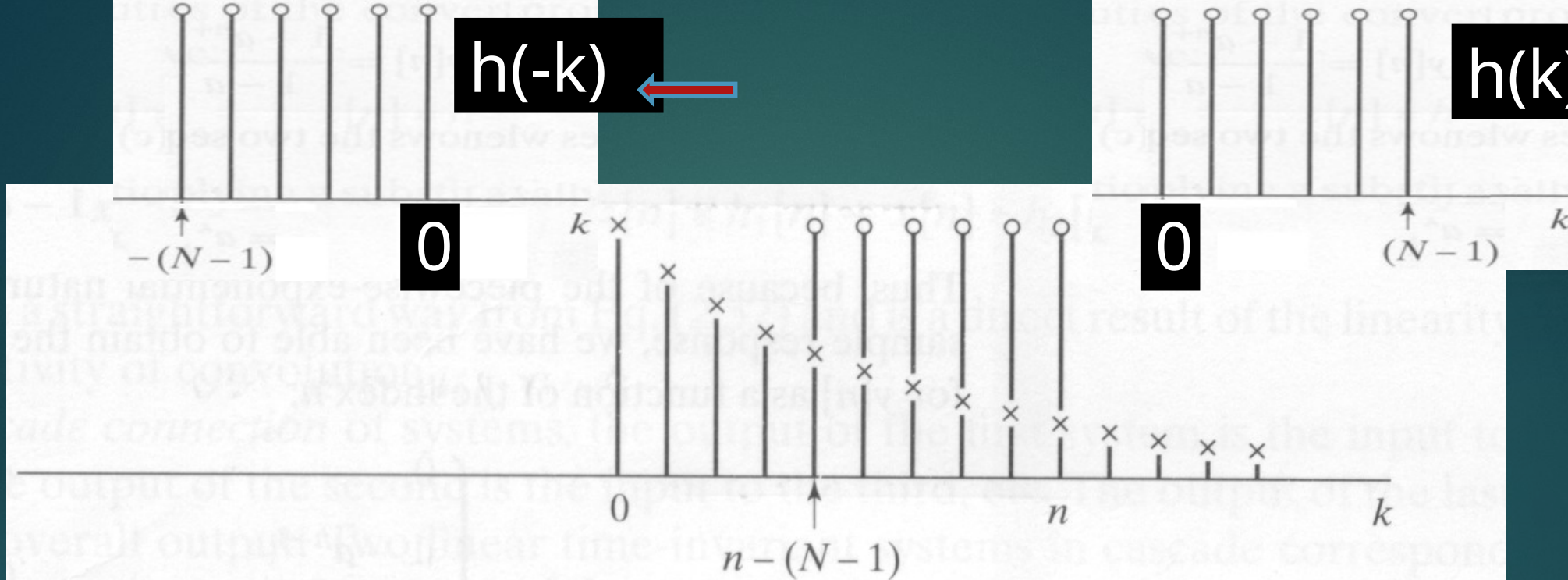


$$y[n] = 0 \quad n < 0$$



$$n \geq 0, \quad n - (N - 1) \leq 0 \quad \Rightarrow \quad 0 \leq n \leq N - 1$$

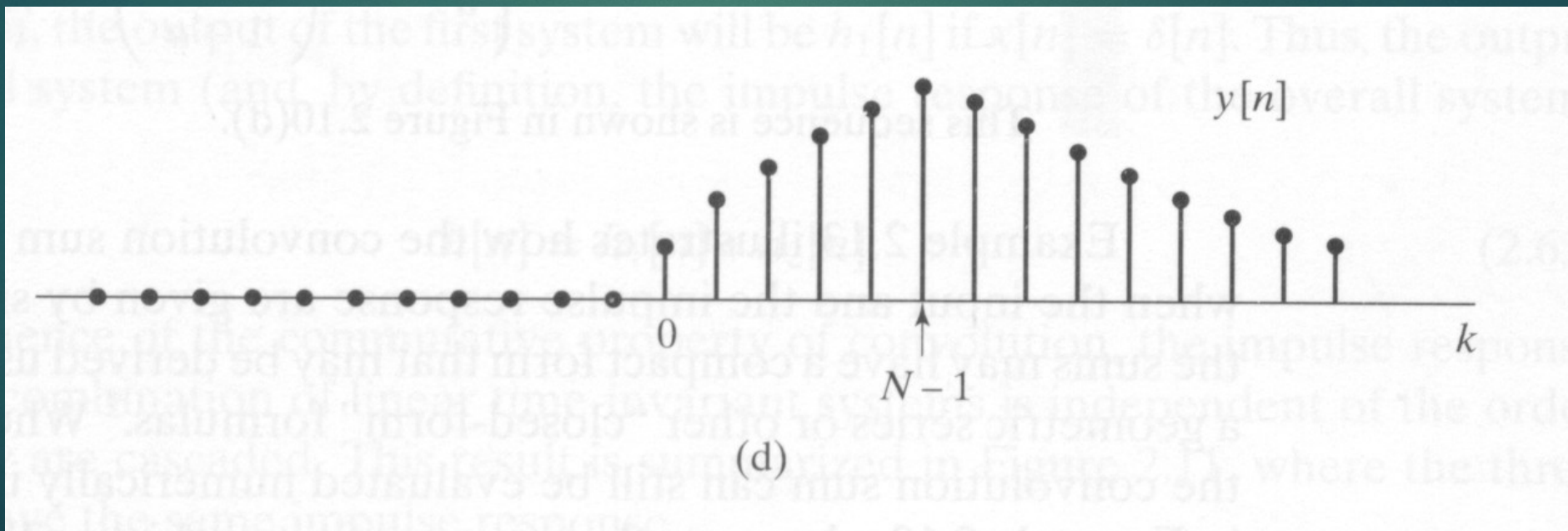
$$y[n] = \sum_{k=0}^n x[k] h[-(k - n)] = \sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a}$$



$$n - (N - 1) > 0 \Rightarrow n > N - 1$$

$$\begin{aligned}
 y[n] &= \sum_{k=n-N+1}^n x[k] h[-(k-n)] = \sum_{k=n-N+1}^n a^k \\
 &= \frac{a^{n-N+1} - a^{n+1}}{1-a} = a^{n-N+1} \left(\frac{1-a^N}{1-a} \right)
 \end{aligned}$$

$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1 - a^{n+1}}{1 - a}, & 0 \leq n \leq N - 1 \\ a^{n-N+1} \left(\frac{1 - a^N}{1 - a} \right), & N - 1 < n \end{cases}$$



Properties of LTI Systems

- ▶ Convolution is **commutative**

$$x[n] \star h[n] = h[n] \star x[n]$$

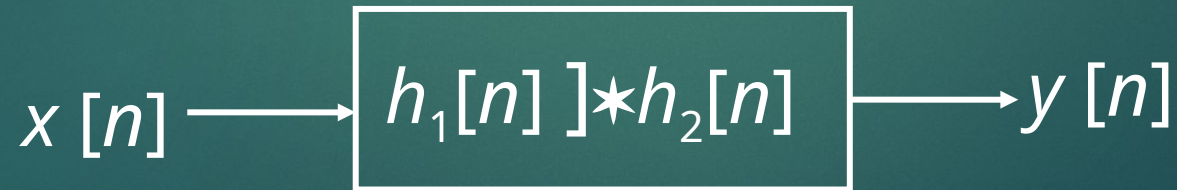
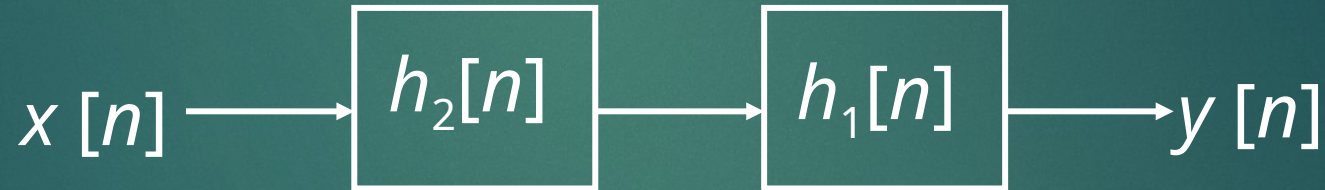
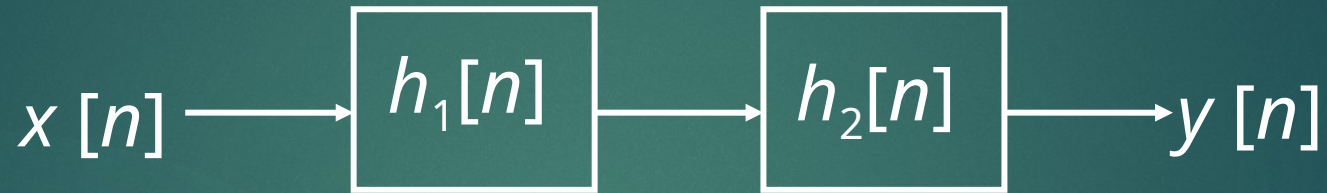


- ◆ Convolution is **distributed** over addition

$$x[n] \star (h_1[n] + h_2[n]) = x[n] \star h_1[n] + x[n] \star h_2[n]$$

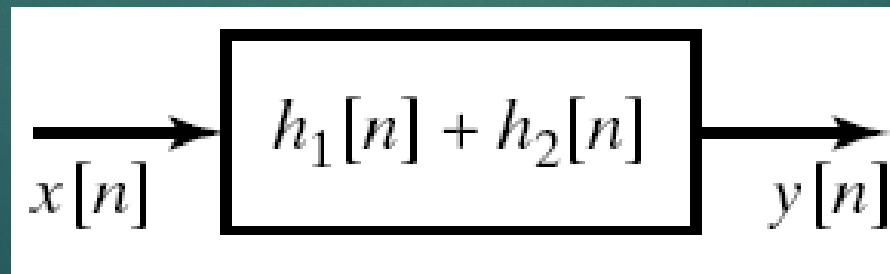
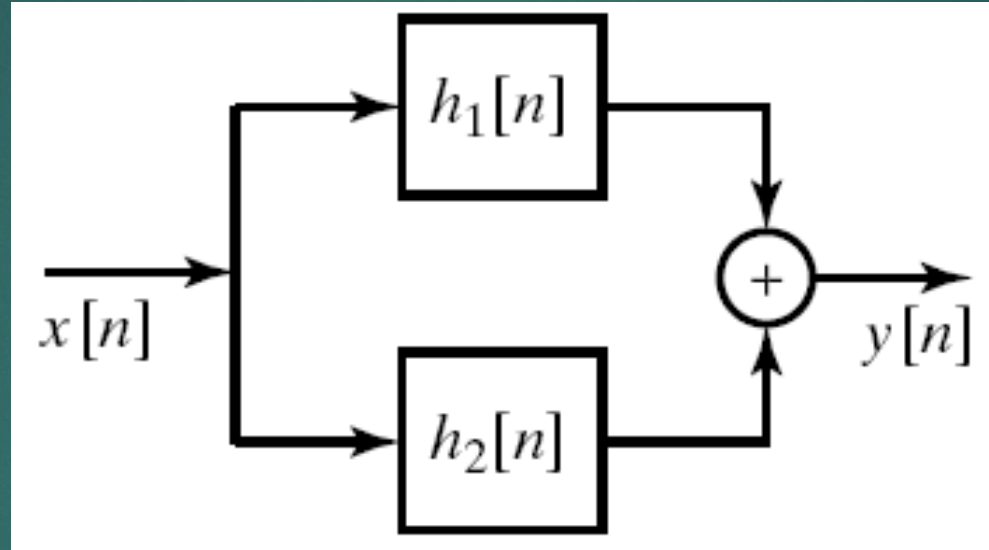
Cascade connection of systems

$$h[n] = h_1[n] \star h_2[n]$$



Parallel connection of systems

$$h[n] = h_1[n] + h_2[n]$$



Stability of LTI Systems

- ▶ LTI system is stable if the impulse response is absolutely summable.

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|$$

$$|x[n]| \leq B_x \longrightarrow |y[n]| \leq B_x \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

Causality of LTI systems

$$h[n] = 0, \quad n < 0$$

Impulse response of LTI

systems

- ◆ Impulse response of Ideal Delay systems

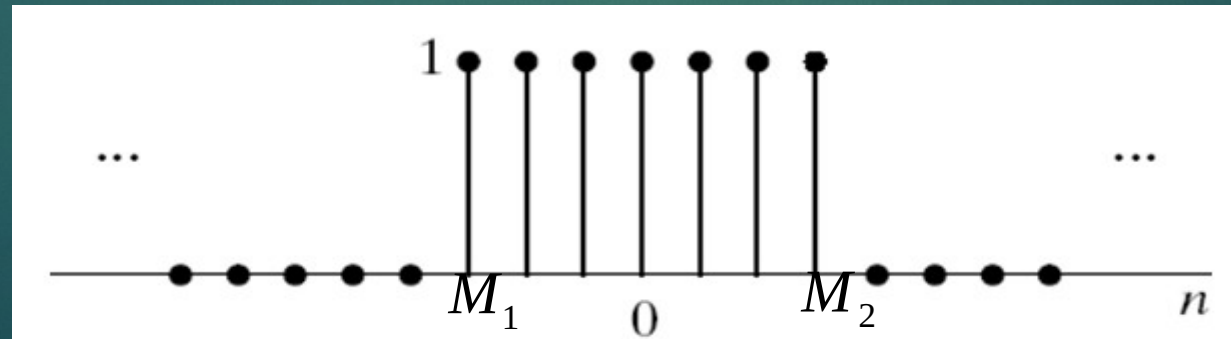
$$h[n] = \delta[n - n_d], \quad n_d \text{ a positive fixed integer}$$

- ◆ Impulse response of Accumulator

$$h[n] = \sum_{k=-\infty}^n \delta[k] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = u[n]$$

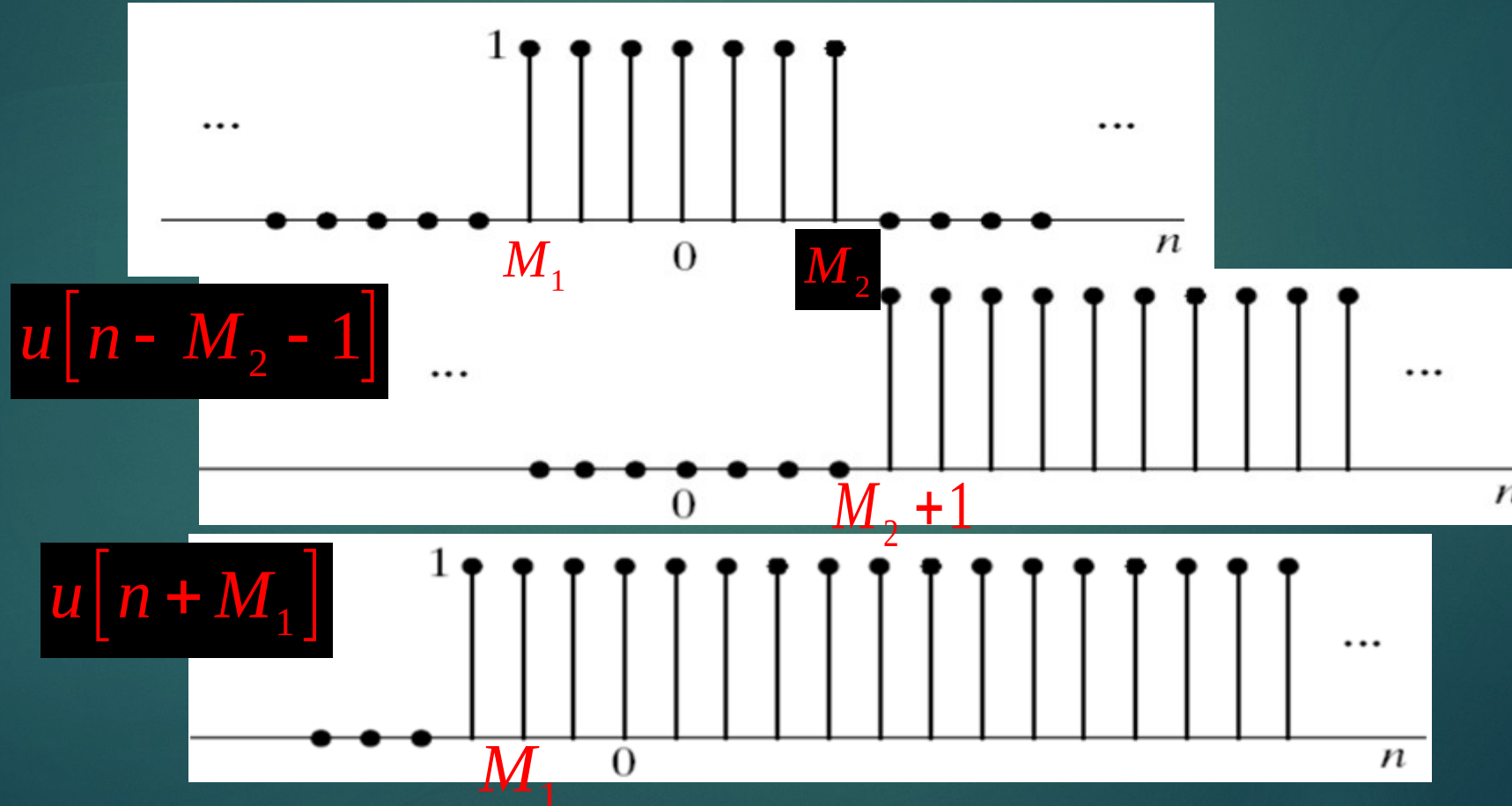
Impulse response of Moving Average systems

$$\begin{aligned} h[n] &= \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} \delta[n - k] \\ &= \begin{cases} \frac{1}{M_1 + M_2 + 1}, & -M_1 \leq n \leq M_2 \\ 0, & \text{otherwise} \end{cases} \\ &= \frac{1}{(M_1 + M_2 + 1)} (u[n + M_1] - u[n - M_2 - 1]) \end{aligned}$$



Impulse response of Moving Average systems

$$h[n] = \frac{1}{(M_1 + M_2 + 1)} (u[n + M_1] - u[n - M_2 - 1])$$



- 
- ▶ Impulse response of **Forward Difference**

$$h[n] = \delta[n+1] - \delta[n]$$

- ◆ Impulse response of **Backward Difference**

$$h[n] = \delta[n] - \delta[n-1]$$

Finite-duration impulse response (FIR) systems

- ▶ The impulse response of the system has only a finite number of nonzero samples.

such
as:

$$h[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} \delta[n - k]$$

$$= \begin{cases} \frac{1}{M_1 + M_2 + 1}, & -M_1 \leq n \leq M_2 \\ 0, & \text{otherwise} \end{cases}$$

- ◆ The FIR systems always are stable.

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Infinite-duration impulse response (IIR)

- ▶ The impulse response of the system is infinite in duration

$$h[n] = \sum_{k=-\infty}^n \delta[k] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = u[n]$$

$$S = \sum_{n=-\infty}^{\infty} |u[n]| = \infty$$

Unstable
system

Stable IIR System:

$$h[n] = a^n u[n]$$

$$|a| < 1$$

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Useful ideal delay system

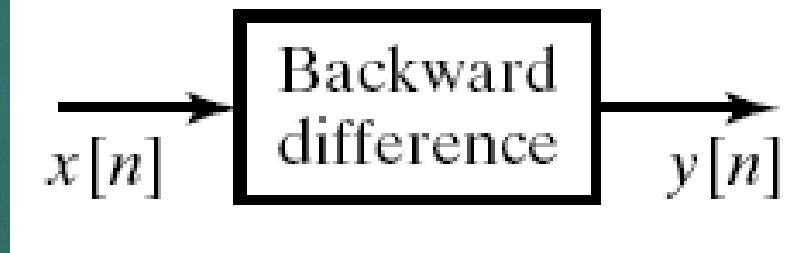
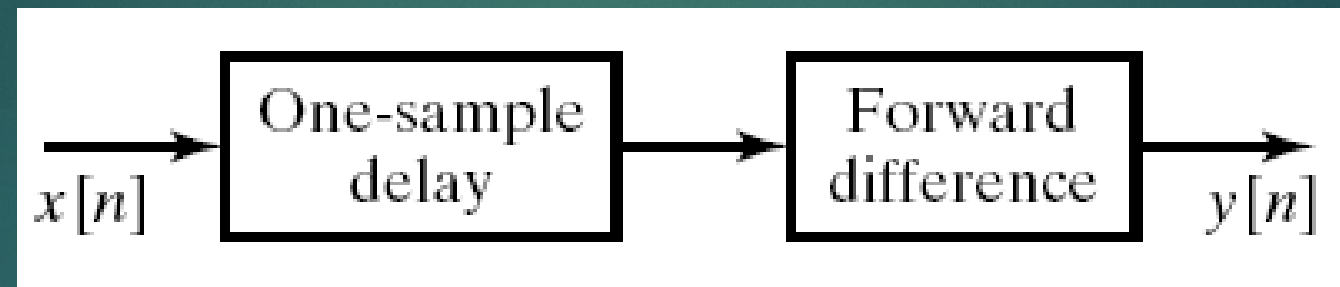
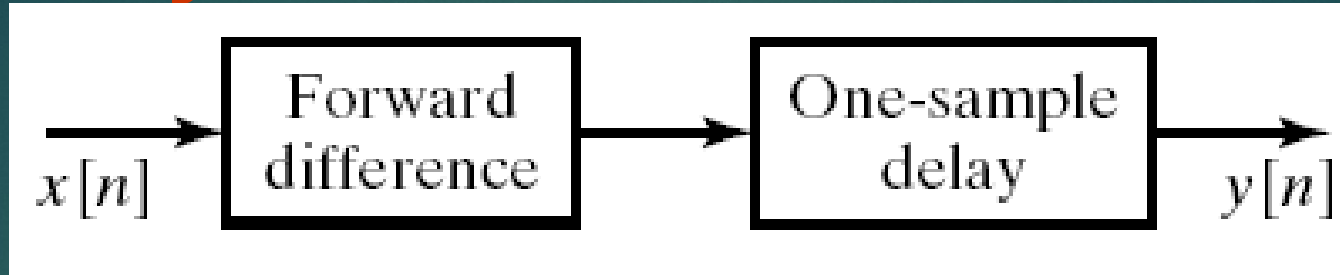
Output of the ideal delay system

$$y[n] = x[n] \star \delta[n - n_d] = x[n - n_d]$$

The convolution of a shifted impulse sequence with any signal $x[n]$ is easily evaluated by simply shifting $x[n]$ by the displacement of the impulse.

Any noncausal FIR system can be made causal by cascading it with a sufficiently long delay.

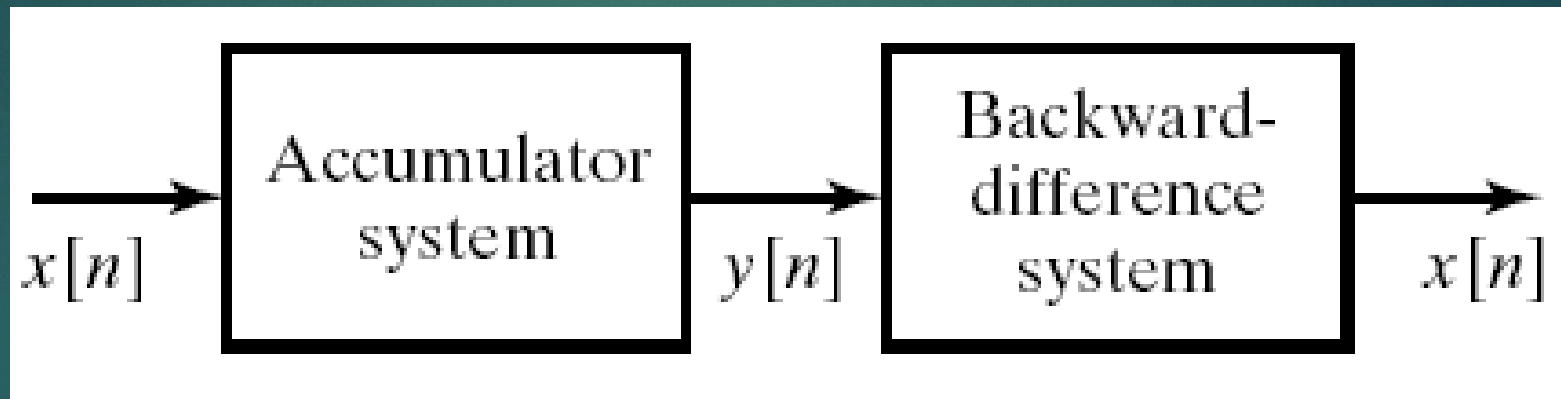
Equivalent systems



$$\begin{aligned} h[n] &= (\delta[n+1] - \delta[n]) * \delta[n-1] \\ &= \delta[n-1] * (\delta[n+1] - \delta[n]) = \delta[n] - \delta[n-1] \end{aligned}$$

Inverse system

$$h[n] \star h_i[n] = h_i[n] \star h[n] = \delta[n]$$



$$\begin{aligned} h[n] &= u[n] \star (\delta[n] - \delta[n-1]) \\ &= u[n] - u[n-1] = \delta[n] \end{aligned}$$